

ProMax® Training

BRE 101 Oil & Gas



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ProMax Oil & Gas Foundations

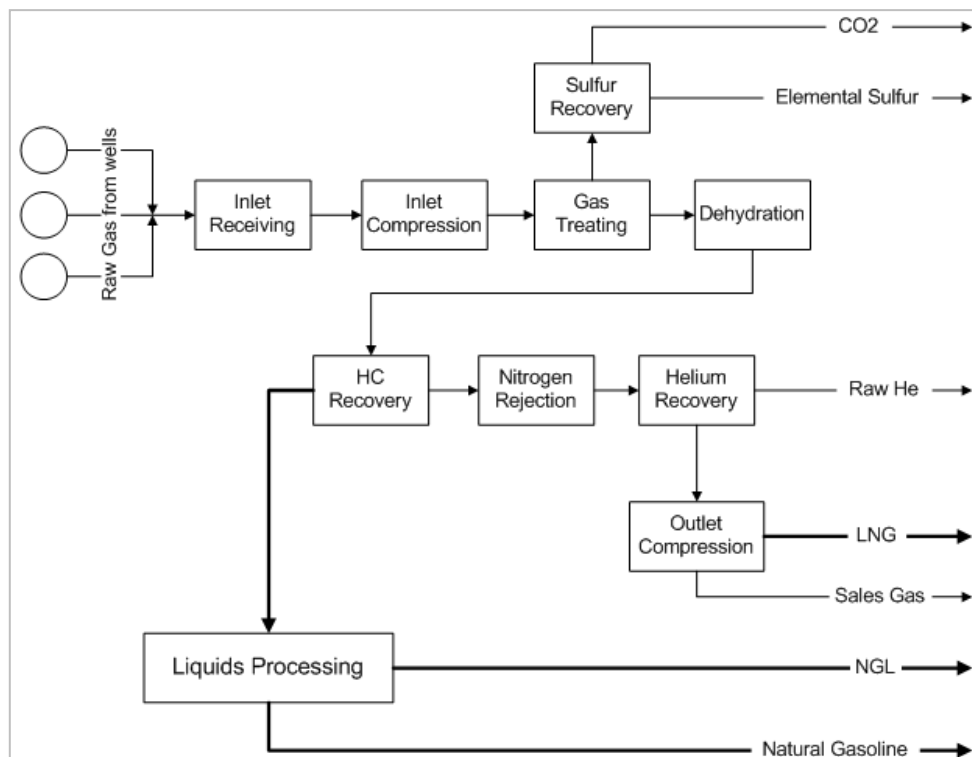
1.1. COURSE DESCRIPTION

ProMax is a flexible, stream-based process simulation package used for the design and optimization of gas processing, refining, and chemical facilities. ProMax provides flexibility to its users through access to over 65 predefined thermodynamic package combinations and over 3200 components, along with crude oil characterization and compound species capabilities. For unit operations, the user has access to pipelines, fluid drivers (compressors and pumps), heat exchangers, vessels, distillation columns, reactors, membranes, and valves.

In addition, ProMax provides OLE automation tie-ins, specifiers, solvers, and Microsoft Excel® spreadsheet embedding, which give the user full access and control of all the information within any stream or block.

The exercises in this course are designed to show the basic functionality of ProMax. This includes the basics of how to draw flowsheets, create Environments, specify processes, and then display information from a completed project. This course also demonstrates many ProMax features for automation and optimization, such as how to set up Simple Specifiers and Solvers and how to utilize the Scenario Tool™. PSV and separator sizing, along with heat exchanger rating, are also discussed. These topics are discussed using applicable oil and gas processing focused exercises.

The general purpose of processing natural gas is to convert a raw gas that leaves the well into saleable products. This process can be separated into 3 main areas: collection and pre-processing, gas treating, and gas processing. ProMax can be used to model most of the systems used in these three areas. The figure below shows a general gas processing flow diagram; however, the processing steps might vary depending on the raw gas composition and requirements. In addition, field processing, such as dehydration and sweetening, may be required before the gas can be sent to the processing facility. Many of these systems are discussed and analyzed within the exercises of this course.



1.2. COURSE AGENDA

Below is a typical course agenda. Note that the agenda may change according to course flow and attendee interests.

DAY 1 Exercise 1: Simple Gas Plant
 Exercise 2: Pipeline Simulation
 Exercise 3: Simple MDEA Sweetening Unit
 Exercise 4: Export/Append Project

DAY 2 Exercise 5: Glycol Dehydration Unit
 Exercise 6: Simple Turboexpander Demethanizer
 Exercise 7: Excel Import/Export
 Exercise 8: Sulfur Recovery Unit
 Exercise 9: Scenario Tool
 Exercise 10: Simple Specifiers

DAY 3 Exercise 11: Simple Solvers
 Exercise 12: User Value Sets
 Exercise 13: Incinerator
 Exercise 14: Separator Sizing
 Exercise 15: Depressurization
 Exercise 16: Shell & Tube Heat Exchanger Rating

Help & Troubleshooting

2.1. PROMAX HELP

Inside the ProMax software, the ProMax Help is the most complete and thorough guide for understanding the ProMax software. It can be accessed by selecting the blue question mark icon in the ProMax Ribbon, or by pressing F1 while inside the Project Viewer.



The ProMax Help contains property definitions, equipment diagrams, and step-by-step instructions for utilizing different ProMax tools and capabilities.

2.2. WEB TUTORIALS

An ever-expanding library of tutorial videos is available online. Topics include explanations of ProMax blocks, how-to demonstrations on different ProMax tools, discussions on commonly simulated processes, and much more! These web tutorials can be found at <https://www.bre.com/Support-Tutorials.aspx>.

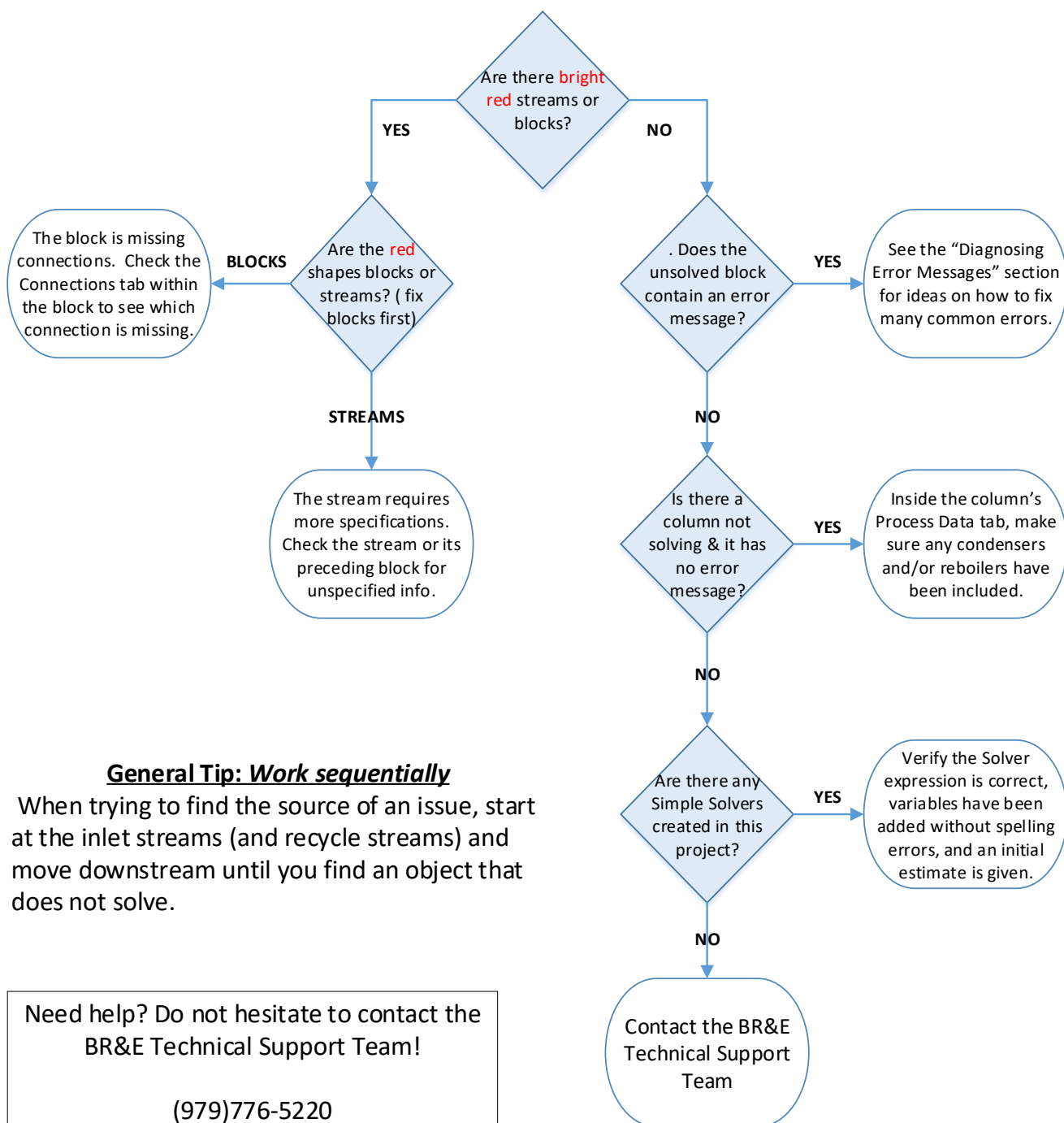
2.3. BASIC SPECIFICATIONS FOR PROMAX BLOCKS

Each block within ProMax gives the user some freedom to specify what is occurring around that block. There are many possibilities of what can be specified within and around each type of block, not all of which are covered here. The following is a list of the most common types of specifications:

- **Inlet Streams** The user must fully specify inlet streams. This generally means providing the following four specifications: Temperature, Pressure, Flow Rate (molar, mass, or volumetric flow), and Composition.
- **Pumps/Compressors/Expanders:** An efficiency and outlet pressure are the most common specifications.
- **Heat Exchangers:** A pressure drop is required within every heat exchanger. For multi-sided exchangers a pressure drop is required for each side. Along with the pressure drop, a single or double-sided heat exchanger allows for one temperature-related specification to be set. Most commonly, one of the outlet temperatures is set, but occasionally a duty or Minimum End Approach Temperature may be set instead.
- **Separator:** A pressure drop or outlet pressure is required. If an energy stream is attached to a separator (such as with a condenser) then an outlet temperature is also specified.
- **Valves:** An outlet pressure (most common) or pressure drop must be specified.
- **Columns:** Columns are allowed one specification for every energy source to the column (Reboiler, Condenser, etc.). The specifications the user chooses to use will vary greatly depending on the process. Regardless of configuration, all columns need a user-specified pressure profile.
- **Process Recycles:** Process recycles allow the user to estimate what is happening at a particular point in the process to allow ProMax to iterate towards the final solution. For these blocks, the recycle outlet stream must be completely user-specified just like an inlet stream (temperature, pressure, flow rate, composition).
- **Q (Energy) Recycles:** Energy recycles require the user to estimate how much energy is being recycled. This energy estimate is user-specified within the block itself.

*Note that operating conditions (temperature, pressure, etc.) are specified in streams. Equipment conditions (efficiency, pressure drop, etc.) are specified in blocks.

2.4. CONVERGENCE HELP GUIDE



2.5. BLOCK & STREAM COLOR CONVENTIONS

ProMax uses color-coding for streams and blocks to help the user along the way, as described by the following table:

Color		Block Status	Stream Status	Comments
RED		Unconnected	Not ready	Stream: specification(s) missing Block: stream connection(s) missing
BLUE		Unsolved	N/A	Block has the min. number of connections required
BROWN		N/A	Unsolved	Stream is fully specified and ready for execution
ORANGE		Approximate	Approximate	Properties inside a stream or block may also be orange, which generally indicates a calculation outside the correlation range
GREEN		Solved	Solved	

2.6. DIAGNOSING ERROR MESSAGES

- ✓ Click the “Warnings” button in the ProMax Ribbon to bring up a list of all the errors and warnings currently in the ProMax project. Users should always check the Warnings list, even if the simulation is fully converged.
- ✓ The Message Log records information about convergence during execution. When troubleshooting, look at the most recent information at the bottom of the Message Log by right-clicking in the log and selecting “Page End”.
- ✓ Most “Errors” prevent a simulation from converging. “Warnings” appear when the simulation has converged but there are certain parts of the simulation that are outside normal operations or normal simulation practices. Thus, the user can determine whether or not a warning needs to be addressed.

Interpreting ProMax Monikers

Every object and property within ProMax has a “moniker” that acts as an identifier for that object (similar to a web address). Error messages include the object moniker which identifies where the error occurred. Every moniker for a stream or block follows this pattern:

ProMax:ProMax!Project!Flowsheets!Gas Processing!PStreams!16

ProMax:ProMax!Project

every moniker begins with this reference meaning it is an object found in the ProMax project

Flowsheets!Gas Processing

the moniker mentions which flowsheet it is looking at (a flowsheet named “Gas Processing” in this case).

PStreams!16

this portion mentions which object the moniker is referring to (a process stream named “16” in this case). PStream stands for process stream. QStream refers to energy stream, and Block refers to any block.

THE FOLLOWING IS A NON-EXHAUSTIVE LIST OF POTENTIAL ERROR MESSAGES

Columns

Under Specified

Inside a column, at the very bottom of the Project Viewer, it will say “Under Specified” when there are degrees of freedom available to the column that have not been used.

Error: Some stage pressures are unspecified. Either the Pressure Change or at least the top and bottom stage pressures must be specified.

This error occurs when the pressure profile has not been configured within the tower. This means that, on the Stages grouping within the Process Data tab, either a pressure change or top/bottom stage pressures need to be set.

Error: The outer loop failed due to an error in the Kb parameters

OR

Error: The outer loop numerical method failed due to the error: Function is oscillating.

OR

Error: The outer loop numerical method failed due to the error: Maximum number of iterations has been reached.

OR

A column turns orange (meaning it is approximating a solution).

These errors mean the column could not find a solution. *First, verify that all the column inlet streams and column specifications are correct.* If these are correct, go to the Convergence tab in the column and change the Enthalpy Model and/or the Inner Loop Model. Sometimes a different model can find a solution when the default did not.

If none of these combinations work, click the “Delete Last Solution” button within the Convergence tab and again try to execute the column. Adding a K Damping factor (0-10) can help with oscillation issues. Checking the “Boston-Sullivan Kb” option or changing the MESH Method can also help with convergence. Lastly, the max number of iterations can be increased by going to the “Solver” grouping under the Convergence tab.

Warning: Model Type Ideal Stage and General Ideal Stage is inconsistent with the electrolytic environment PR Amine Environment and the component list containing CO2 and an ethanolamine. It is recommended that you select Model Type Ideal Stage and TSWEET Kinetics

This warning occurs when an amine absorber is run as the incorrect column type. On the Process Data tab in the Column grouping, make sure “TSWEET Kinetics” is selected for the Ideal Stage Column Type.

Warning: Model Type Ideal Stage and General Ideal Stage is inconsistent with the electrolytic environment PR Amine Environment and the component list containing CO2. It is recommended that you select Model Type Ideal Stage and TSWEET Stripper.

This warning occurs when an amine stripper is run as the incorrect column type. On the Process Data tab in the Column grouping, make sure “TSWEET Stripper” is selected for the Ideal Stage Column Type.

Error: The bottom of the column has no Vapor feed

OR

Error: The top of the column has no Light Liquid feed

All columns must have vapor entering on the bottom stage and liquid entering the top stage in order to converge.

Error: A duty estimate is required for the pumparound loop exiting from PStream 4 and returning via PStream 8

This error message means a duty estimate is required for a column pumparound loop. If duty is added to a pumparound by a heat exchanger (other than the condenser) or pump, an estimate may be required for the total pumparound duty amount. This estimate can be made within the column by adding a "Pump-around Estimate" on the Specifications tab. Estimate a negative duty if energy is being removed from the loop.

Heat Exchangers

Always check inside the exchanger to make sure a Pressure Drop is specified. If it is a double-sided exchanger, then it will require a Pressure Drop on both sides of the exchanger.

Error: Exchanger Gas/Glycol HEX - A contains a negative or close endpoint approach temperature. Rating is not possible.

This error message means there is a temperature cross in the exchanger, indicating that the specified heat transfer is thermodynamically impossible. While the exchanger will converge, any attempt to rate it will fail. Adjust any duty or outlet temperature specification until there is no longer a negative approach temperature for the exchanger.

Error: Block is overspecified.

This error means the exchanger is overspecified. Most likely either both exchanger outlet temperatures have been specified or one outlet temperature plus a Minimum Approach Temperature have been specified.

Pumps

Warning: Efficiency out of range.

This error occurs when ProMax calculates a pump efficiency greater than 100%. This usually occurs when the user specifies an outlet temperature from the pump. Generally, it is recommended to not specify the pump outlet temperature; rather, specify an efficiency inside the pump and allow ProMax to calculate the outlet temperature.

Error: Block is overspecified.

This error means the pump is overspecified, likely due to both a pump outlet pressure and temperature being specified. Generally, it is recommended to not specify the pump outlet temperature; rather, specify an efficiency inside the pump and allow ProMax to calculate the outlet temperature.

Recycles

Error: Maximum number of iterations has been reached

This error message means that the recycle block was not able to converge to a solution after performing the set maximum number of iterations. The first thing to do is verify that there is a reasonable estimate in the recycle block outlet stream. The maximum number of iterations can be increased inside the Recycle block under the Solver grouping on the Converge tab.

Simple Solvers

 Error number: 80020101
Source: ProMax:ProMax!Project!Calculators!Air Flow Solver

Script Error: 'TailgasRatio' is undefined at Line 1 and Column 0.
"Microsoft JScript runtime error"

This message indicates that a variable written into the Solver/Specifier equation is undefined. This means either *a*) the variable has not been added to the Measured/Independent Variable list or *b*) the name written in the equation does not match the name given to the Measured/Independent Variable. Make sure the variable is listed and that the names match, keeping in mind that names are case-sensitive and no spaces are allowed in a variable name.

 Error number: 800432E9
Source: ProMax.ProMax.1

The PropertySolver of CalcVariable AFS for the Calculator Air Flow Solver does not have a valid guess.

This message indicates that a Solver initial guess has not been set. Inside the solver, make sure an initial estimate has been entered into the Calculated Value cell. ProMax will adjust this initial guess until it reaches a solution.

Solver/Recycle Priorities

When there is more than one solver and/or recycle in a simulation, priorities tell ProMax the order in which to solve the different solvers/recycles. ProMax will solve from the highest priority found in the simulation on down. Poor priority order can lead to significantly longer convergence times and even prevent a simulation from solving. Consider the following:

- Give higher priorities to solvers that come first in sequence. Example: solvers/recycles on an amine unit should solve before (and therefore have higher priorities than) solvers/recycles on a downstream dehydration unit.
- Nested, inner loops should be solved before outer loops. Example: solvers within a recycle loop should solve before the loop's recycle block.
- The exact number value of a priority is unimportant; it is only the comparison of priorities from highest to lowest that matters. Example: if a recycle has a priority of 10 and a solver has a priority higher than 10 (11, 20, 100, etc.), ProMax will solve the solver and then the recycle.
- A list of solvers/recycles in a project, including their priorities, is found in the Solver Summary within the Project Viewer.

Streams

Error: Flash Calculations have failed or an illegal property exists.

This error means that either the calculated temperature or pressure of a stream is outside the max/min allowable values. The most common cause is a duty in an exchanger that is impossibly high for the given stream's flow rate.

Warning: The temperature of 54.867 °F is within 10 °F of hydrate formation.

OR

Warning: The temperature of -54.3304 °F is below hydrate formation.

These warnings appear for any stream that is near or below a temperature where hydrates could form. *Similar warning messages appear for ice and dry ice formation.* This means that solids could potentially form at this part of the process and the user should consider running at higher temperatures, removing water (or CO₂ for dry ice) from the system, or injecting a hydrate inhibitor such as methanol or ethylene glycol.

2.7. VISIO HOTKEYS

Microsoft hotkeys are available within any ProMax project. In addition, users can create custom hotkeys through VBA programming. Below is a list of commonly used Visio hotkeys (based on Visio 2016):

Ctrl + A	Select All
Ctrl + D	Duplicate Selected Shape (creates a new, blank shape)
Ctrl + C Ctrl + V	Copy and Paste Selected Shape(s) (user-defined properties in a stream will be copied along with the stream but internal block info will not be copied with the block)
Ctrl + H	Flip Shape Horizontal
Ctrl + J	Flip Shape Vertical
Ctrl + S	Save File
Ctrl + 3	Toggle to/from Connector Mode
Ctrl + L	Rotate Selected Shape Counterclockwise
Ctrl + R	Rotate Selected Shape Clockwise
Ctrl + Click	Select Multiple Shapes
Ctrl + Wheel	Zoom In / Out
Shift + Wheel	Scroll Flowsheet Right / Left
Shift + Arrow	Nudge Selected Object Right/Left/Up/Down

Course Exercises

Exercise 1: SIMPLE GAS PLANT

Process Background: Collection and Pre-Processing

Natural gas liquids (NGLs) recovered through natural gas processing are often stored until they are trucked or piped away for processing. To avoid excessive emissions and losses, the liquid products may be “stabilized”. The stabilization process heats the liquids to allow the most volatile components to vaporize, lowering the vapor pressure of the remaining NGL product.

Note that for all exercises the alternative specifications in metric units have been rounded for convenient values and are therefore not exact unit conversions.

Goal and Information

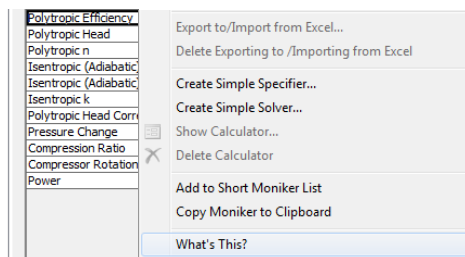
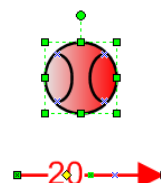
This simple system models a gas feed separated into a liquid product and a sales gas by compressing, cooling, and flashing the gas. The liquid is sent to a stabilizer to achieve product specifications by removing lighter components in the stabilizer overheads. The column is operated with a reboiler that is used to control the composition of the bottom liquids.

- Configure the Environment by selecting an appropriate property package (Peng-Robinson or SRK) and adding the listed components.
- Use the stream conditions in the table to define the inlet stream.
- The compressor outlet pressure is 725 psig [50 barg]. Use 75% polytropic efficiency.
- The outlet of the air cooler should be 110°F [43°C]. A pressure drop should be set in the cooler.
- Sales gas is sent through a gas/gas exchanger to cool the inlet stream. The Minimum End Approach Temperature in this first cooler is 30°F [16°C]. Two pressure drop specifications are required for a two-sided exchanger.
- The second cooler chills the stream feeding the propane cooler to 36°F [2°C]. Set the pressure drops.
- The propane refrigeration cools the stream entering the low temperature separator to 20°F [-6.5°C]. Set a pressure drop.
- Set a pressure drop of 0 psi in the LTS.
- The pressure at the outlet of the JT valve is 200 psig [14 barg].
- The column has 8-stages with a 4 psi [0.28 bar] pressure change.
- Common NGL product specifications include: C1 < 0.5 liquid volume percent (LV%) of the NGL product stream; and a C1/C2 LV% ratio < 0.015. Inside the stabilizer on the Specifications tab, create both of the specifications mentioned here (see the FAQ on the next page for assistance). Set the C1/C2 ratio as a Specification and leave the C1 LV% as a Calculation.

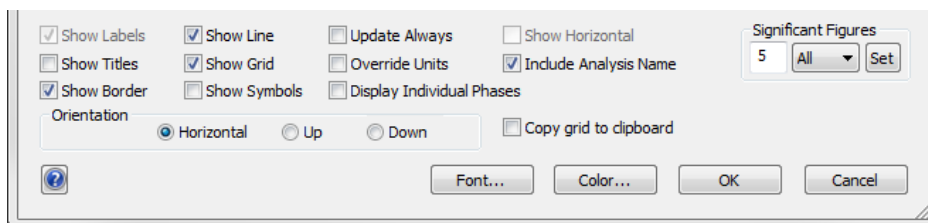
Inlet Conditions		
Temperature	95°F	[35°C]
Pressure	200 psia	[14 bar]
Flow	5 MMSCFD	[5600 Nm ³ /h]
Composition		Mole %
Nitrogen		1.92
Carbon Dioxide		2.71
Methane		58.66
Ethane		16.17
Propane		12.86
i-Butane		1.58
n-Butane		0.70
i-Pentane		0.84
n-Pentane		3.73
n-Hexane		0.81

FAQs on Exercise 1

- All heat exchangers normally have a pressure drop specified, since this is a property of the exchanger. Setting an outlet pressure instead can result in unexpected results if conditions change later.
- Generally, an efficiency is set for all pumps, compressors, and expanders. Setting both the outlet pressure and temperature is strongly discouraged because a small inaccuracy in the temperature can result in efficiencies less than 0% or greater than 100%.
- Visio shapes can be resized and rotated by utilizing points that become available when the object is selected. Process and Energy streams can be redirected from similar points.
 - The small green circle above the exchanger to the right is for rotating the shape; a few shapes restrict this to maintain correct positioning of vapor and liquid exits.
 - The other green squares are to resize and stretch the shape
 - The yellow diamond, as show on the “20” in the stream to the right, designates label location; labels can be moved by dragging this diamond.
 - To redirect streams, find the small bright-green squares located in the middle of the streams, then click and drag from these points.
- When the definition of a property in ProMax is unknown, often the “What’s This” Help menu can provide some assistance. This feature is found by right-clicking on most cells in the Project Viewer:



- The Callout box option is found in the “ProMax Streams” set, and contains several set-up options to customize the display. The “Font...” button gives control over the font size, color, and type. The “Color...” button allows changes to the gridline colors.



- For the Column Specifications mentioned, use “Component Flow” for C1 in total bottoms liquid stream and “Component Ratio” for the methane to ethane ratio.
- Recycle blocks always require the outlet PStream to be completely specified (Pressure, Temperature, Flow Rate and Composition).
- Many additional stream properties (such as the Phase Envelope, Vapor Pressure, Line Sizing, etc.) are calculated within the different analysis available in a stream under the Analysis tab. For a comprehensive list of analysis available in ProMax, see page the ProMax Help.

Exercise 2: PIPELINE SIMULATION

Goal and Information

Two different natural gas lines are mixed and sent through 17.3 miles of pipeline to a processing facility. Along the way, a 3rd gas stream from a different production facility is added to the mixture. Build a simulation to represent this system based on the information below.

PIPE-1:

Segment 1: ▶ 24,000 ft [7315 m] length
▶ 8 in [200 DN], schedule 80
▶ Ambient T = 50°F [10°C]

Segment 2: ▶ 45° standard elbow

Segment 3: ▶ 25,000 ft [7620 m] length
▶ 8 in [200 DN], schedule 80
▶ 500 ft [150 m] elevation rise
▶ Ambient T = 40°F [5°C]

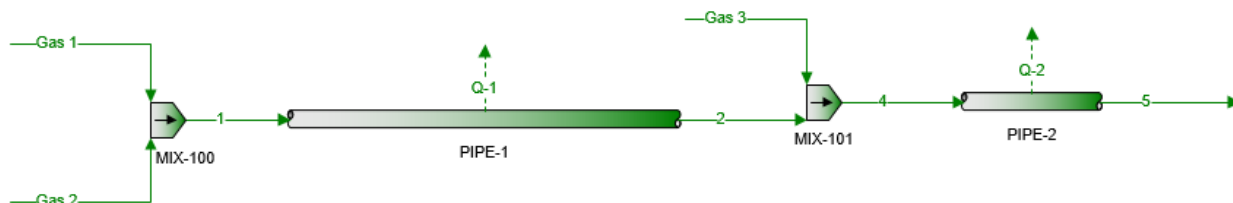
Segment 4: ▶ 30,000 ft [9140 m] length
▶ 8 in [200 DN], schedule 80
▶ Ambient T = 30°F [-1°C]

PIPE-2:

Segment 1: ▶ 2.3 mile [3700 m] length
▶ 10 ft [3 m] elevation rise
▶ 12in [300 DN], schedule 80
▶ Ambient T = 40°F [5°C]

Stream	Gas 1	Gas 2	Gas 3
Temperature	100°F [37°C]	100°F [37°C]	70°F [21°C]
Pressure	600 psig [41.5 barg]	600 psig [41.5 barg]	510 psig [35.2 barg]
Flow rate	8.2 MMSCFD [9153 Nm ³ /h]	5 MMSCFD [5575 Nm ³ /h]	8 MMSCFD [8930 Nm ³ /h]
Composition	Mole %		
CO ₂	0.2	0.3	0.1
N ₂	0.5	0.3	0.8
C1	84.3	86.9	85.1
C2	5.5	4.8	6.5
C3	5.1	4.1	3.8
iC4	1.4	0.9	1.1
nC4	1.5	1.3	1
iC5	0.85	0.7	0.6
nC5	0.3	0.3	0.5
nC6	0.3	0.2	0.37
nC7	0.05	0.2	0.13

- The heat transfer coefficient should be calculated by ProMax based on Carbon Steel A134 welded pipe with a centerline buried depth of 24 inches [600 mm] in dry clay ground.



QUESTIONS

- What is the temperature and pressure of the gas exiting the pipeline to the processing facility?
- What is the calculated overall heat transfer coefficient for the first pipe segment?
- For each pipe segment, plot length vs. pressure.
- Where does drip condensate begin to form? How much liquid (bbl/d) exits Pipe-2?
- If the upstream dehydration facility shuts down and Gas 1 becomes saturated with water, do hydrates form?

Exercise 3: SIMPLE MDEA SWEETENING UNIT

Process Background: Acid Gas Removal

The removal of acid gases, specifically hydrogen sulfide (H_2S) and carbon dioxide (CO_2), can be achieved through several methods, including the use of: solid scavengers (e.g. iron sponges, etc.), chemical solvents (e.g. MEA, DGA, MDEA, caustic, potassium carbonate, etc.), physical solvents (e.g. MeOH, DEPG, NMP, etc.), and membranes. Some recently developed processes reduce the H_2S in the natural gas directly to solid sulfur using liquid catalysts or biological pathways. The chosen method depends on the temperature, pressure and inlet gas composition.

An example of a chemical solvent process can be found in this exercise. Further examples of chemical and physical solvent processes can be found in the example projects located at *File → Open Example ProMax Project*. The chemical solvent examples are contained in the “Amine Sweetening” directory and the physical solvent examples can be found in the “Gas Processing” directory. See the ProMax Help topic for information on modeling membranes.

Chemical solvents

Chemical solvents usually consist of one or more amine compounds mixed with water. The solvent is brought into intimate contact with the gas in an absorption column that typically has ~20 real trays or equivalent packing. It is common practice to cool the temperature of the lean solvent entering the absorber to about 10°F [5°C] above the inlet gas temperature since colder temperatures improve absorption.

In the absorber, the acid gases are absorbed into the solvent and react with the amine (a weak base). This process is kinetically limited. Primary amines (e.g. monoethanolamine (MEA), diglycolamine (DGA), etc.) have the fastest kinetics, secondary amines (e.g. diethanolamine (DEA), diisopropanolamine (DIPA), etc.) and tertiary amines (e.g. methyl-diethanolamine (MDEA)) have respectively slower kinetics. ProMax accounts for kinetic effects with a rigorous rate-based kinetic model.

While primary and secondary amines can absorb H_2S and CO_2 quickly, they are typically used at lower concentrations and with a reclaimer to reduce the amount of amine loss and degradation. Tertiary amines can be used at higher concentrations and the reaction kinetics favor selective absorption of H_2S over CO_2 . Typically, these tertiary amines have no reclaimer since their degradation and losses are much lower.

If the absorber is operated at high pressure, the rich amine is sent through a flash tank to reduce the amount of hydrocarbons present in the rich amine before it reaches the regenerator. A lean/rich heat exchanger is also commonly used to reduce reboiler duty.

The regenerator is a distillation tower that typically has ~20 real trays. It is operated at the lowest pressure feasible for downstream processing of the acid gas, typically near 15 psig [2.05 bara]. The reboiler often uses saturated steam at 50 psig [4.5 bara] and a flow rate equivalent to 0.7-1.5 lb steam/gal [90-180 kg steam/ m^3] of circulating solvent. A condenser is also present to reduce the amount of amine loss and to reduce the amount of water sent down-stream.

Additional information on optimization of sweetening systems can be found in the “BRE 231: Sour Gas Processing” course or within the ProMax Help.

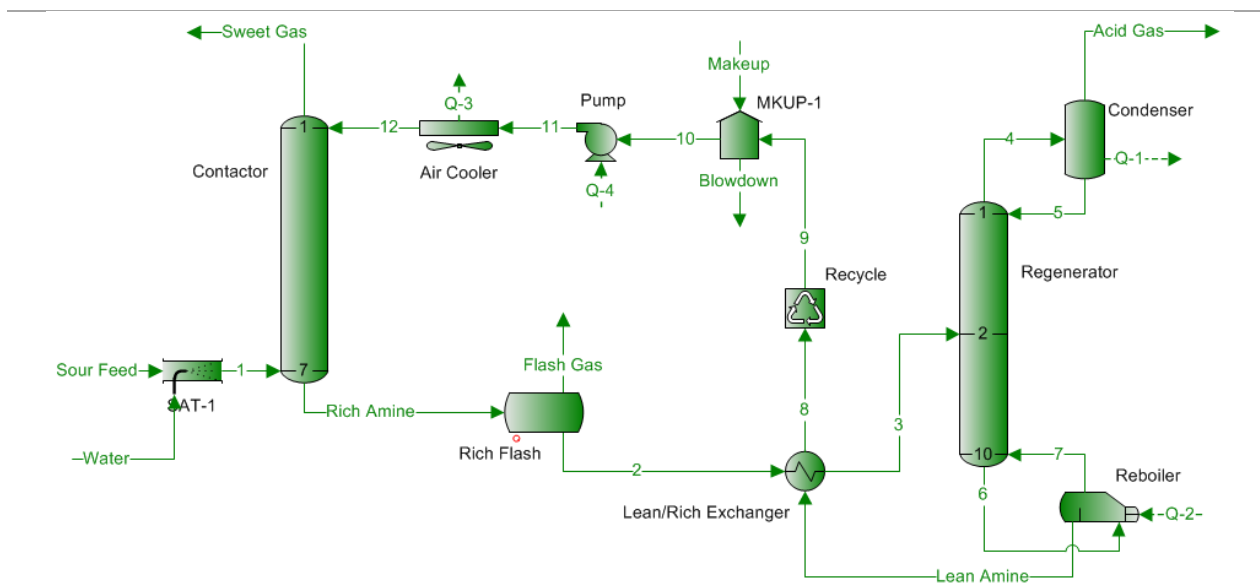
Goal and Information

In this exercise, a sweet gas is required to have less than 2 mol% CO₂ and less than 4 ppm H₂S (molar). Use the following information to simulate this amine sweetening unit and determine if the requirements are being met. Use an Amine Sweetening property package, as there are ionic reactions occurring. Use the “Exercise 3 – Simple MDEA Sweetening Unit.pmx” file.

Part A: Ideal Stage Model

- Select an appropriate Amine Sweetening Environment.
- The inlet gas stream is 100% saturated with water
- Supply a guess for the recycle block outlet.
- 40 wt% MDEA is circulated at 190 sgpm [44 m³/h]. The makeup stream is at ambient temperature and 20 psig [1.4 barg].
- The pump increases the lean amine pressure high enough to return to the absorber column.
- The air cooler outlet temperature is set to 10°F [5°C] warmer than the gas feed temperature.
- The absorber uses the “TSWEET Kinetics” column type that has a 4 psi [0.3 bar] pressure drop.
- The absorber diameter is 4 ft [1.3 m]. Set a system factor of 0.8.
- Set the real/ideal stage ratio to 3. This represents the fact that 3 real amine absorber trays are equivalent to 1 ideal stage (amine absorbers are about 33% efficient).
- Weir height is 3 in [7.6 cm] and tray spacing of 2 ft [0.6 m].
- The rich flash tank operates at 100 psia [6.9 bar].
- The rich amine enters the regenerator at 210°F [99°C].
- The regenerator top stage pressure is 12 psig [0.82 barg] with a 4 psi [0.3 bar] pressure change through the tower.
- Remember to use the “TSWEET Stripper” column type.
- The condenser operates at 120°F [50°C] and the reboiler duty is 17 MMBtu/h [5000 kW].

Inlet Conditions	
Temperature	100°F [38°C]
Pressure	1000psia [70 bar]
Flow	50 MMSCFD [56000 Nm ³ /h]
Composition	Mole %
H ₂ S	1.5
CO ₂	3.5
Methane	83.4
Ethane	5.65
Propane	4.07
n-Butane	0.9
n-Pentane	0.5
n-Hexane	0.32
n-Heptane	0.15
Benzene	0.005
Toluene	0.002
Ethylbenzene	0.001
o-Xylene	0.002



QUESTIONS

1. What are the sweet gas CO₂ and H₂S compositions?
2. What are the liquid residence time and percent flood on an absorber tray?
3. What are the H₂S, CO₂, and total rich loadings (mol/mol)? Lean loadings?
4. What are the lean and rich approach in the absorber for H₂S?

Part B: Trayed Tower Using Mass + Heat Transfer Model

One advantage of the Mass + Heat Transfer Model available in ProMax is the ability to directly specify the number of trays in the column.

- Change the absorber tower to use the “Mass + Heat Transfer” model, and select the “TSWEET Absorber/Stripper” column type.
- Increase the column stages to 20. All column internals (diameter, tray spacing, etc.) remain the same.
- Solve the flowsheet again. How do the results compare?

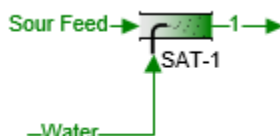
Optional Extended Exercise: Activated MDEA

Activated MDEA is an MDEA/Piperazine mixture, where Piperazine is added to improve CO₂ pickup.

- Add Piperazine to the environment, then modify the makeup block to maintain 3% Piperazine & 40% MDEA.
- How does the H₂S and CO₂ absorption compare? Why does piperazine cause the results to shift this way?

FAQs on Exercise 3

- The Saturator block is used in this exercise to fully saturate the sour feed stream with water. The “Water” stream requires only that a composition of 100% water is set. Allow ProMax to calculate other properties. The default setting for the block is to saturate the process stream to 100% saturation.



Connections		Process Data	Streams	Notes
Fraction of Saturation		100	%	
Outlet Saturant Concentration		58.5844	lbm/MMSCF	

- The Lean and Rich Approach are column specifications used to determine the best way to improve absorber performance. They are also used to predict absorber stability during process upsets.
 - Lean Approach: Analyzes the driving force at the top of the absorber. If the lean approach is too high (>80%) then the lean amine acid gas concentration feeding the tower is high and there will be little driving force to move acid gas into the amine at the top of the column. Increasing the reboiler duty will lower the acid gas concentration in the lean amine and thus increase the driving force at the top of the column.
 - Rich Approach: Analyzes the driving force at the bottom of the column. If the rich approach is too high (>80%) then the amine solution will approach equilibrium with the incoming feed and there will be little driving force to move acid gas into the amine at the bottom of the column. Increasing the amine flow rate will increase the capacity for pickup and increase the driving force at the bottom of the column.
- The Make-up / Blow-down block controls the composition and flow rate of any make-up stream required. The Make-up stream requires a user-set temperature and pressure.

Exercise 4: EXPORT/APPEND PROJECT

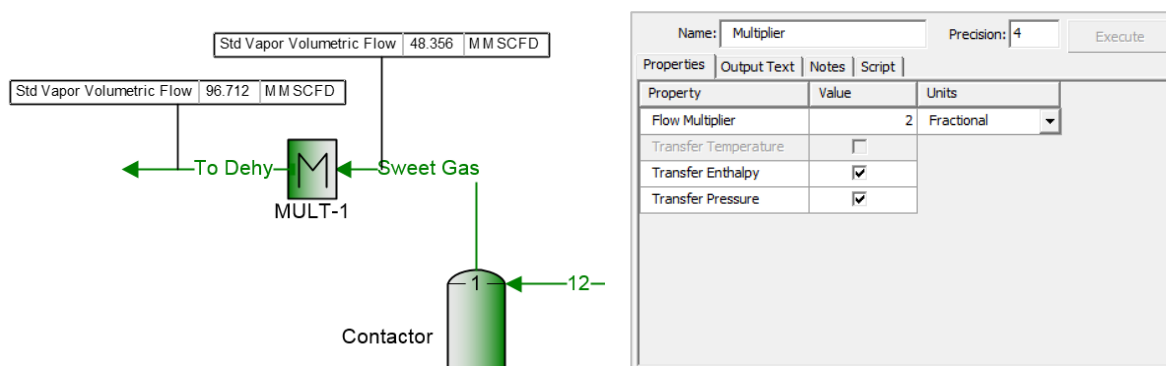
ProMax provides a simple method to combine different projects. The options may be found in the File menu as “Export Project” and “Append Project”. In this exercise, an export file of the amine unit from Exercise 3: is created and appended to an existing template project of a dehydration unit.

PROCESS INFORMATION

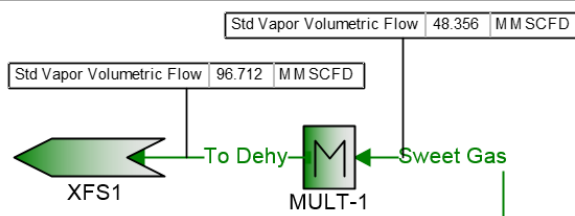
- From the file menu, select “Export ProMax Project...” Save the project in an easily accessible location, and close the file.
- Open the file named “Ex5 - TEG Unit.pmx”.
- From the file menu, choose “Append ProMax Project...” Browse and select the “.pmxexp” file from step 1.
- From the TEG flowsheet, select “Active Flowsheet”. Then Duplicate the MDEA Environment and rename the resulting Environment “TEG Environment”. Assign the property package “SRK” to the Environment, and leave the components unchanged for now. Select OK, and then assign the “TEG Environment” to the TEG flowsheet.

Flow Multiplier

On the MDEA flowsheet, to double the flow rate of the sweet gas, use the ProMax Property Stencil named “Inline Flow Multiplier”. Doubling the flow rate can be used to represent having two identical amine trains that mix before further processing. Allow this shape to propagate the pressure and enthalpy and set the outlet flow rate as double the inlet flow rate. The composition will always be unchanged when using this block. The Property Stencil set is found in the “More Shapes” menu, if not already available in the project.



Drop a Cross Flowsheet Connector, found in the “ProMax Streams” stencil, onto the MDEA page near the Inline Multiplier outlet stream. Select the TEG flowsheet as the connected page. Connect the Inline Multiplier outlet stream to the inlet of the Cross Flowsheet Connector as shown below.



On the TEG flowsheet, connect the second side of the Cross Flowsheet Connector (which appears in the lower left-hand corner of the flowsheet by default) into the dehydration unit “Wet Gas” stream.

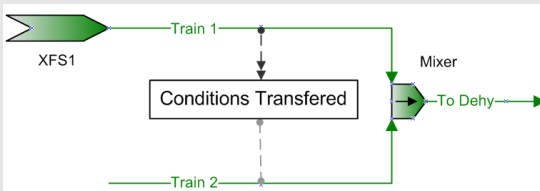
QUESTIONS

1. Is there a temperature difference between the “Sweet Gas” stream in the MDEA flowsheet and on the TEG flowsheet? Why? How can this be corrected? (Incorporate this change into the project.)
2. Since the component list from the MDEA unit earlier was duplicated, the current TEG Environment is lacking the component TEG. When adding TEG to the Environment, also delete the MDEA from the list. Why will the cross-flowsheet connector no longer solve?
3. In the Cross Flowsheet Connector, a Mole Fraction Transfer Threshold can be set to prevent the transfer of trace components across flowsheets. This can be useful in cases where the user wants to remove a trace component from the components list across different units. In this exercise, set the Threshold value to limit transfer to only those components above 5 ppm. What components will not be transferred to the Dehy Flowsheet? What issues does this raise? Change the Threshold value to 1.5 ppm.

 **Tip**

 **Copy Stream Conditions**

The “Copy Stream Conditions” property stencil is an alternative that will display both inlet streams. The shape connects to a reference stream and a blank target stream with connector lines, as illustrated. Composition, mass flow rate, molar enthalpy and pressure are transferred.



Exercise 5: GLYCOL DEHYDRATION UNIT

Process Background: Glycol Dehydration

Water content of a gas stream often needs to be adjusted to meet specifications for transmission pipelines, storage, or further processing. There are two methods currently employed in most dehydration systems: direct injection or a contactor column. Ethylene glycol (EG) and methanol are typically used in the injection method, and triethylene glycol (TEG) is typically used with the contactor column.

A contactor column with packing or trays is used to allow intimate contact between the wet gas and glycol. These towers typically have 2-3 ideal stages. Lean TEG is fed to the top of the tower and contacts the vapor stream coming up from the bottom. The TEG preferentially absorbs water from the gas stream. This rich TEG is then regenerated by removing the water in a regenerating column. The regenerating column typically consists of a very large reboiler with a small packed tower.

In general, about 2-5 gallons of TEG are circulated for every pound of water in the wet inlet gas. The reboiler temperature should not exceed the glycol degradation temperature, approximately 400°F [204°C] for TEG systems. It is also common practice to cool the temperature of the lean glycol entering the contactor to about 10°F [5°C] above the inlet gas temperature since colder temperatures improve absorption.

Though glycols do preferentially absorb water, aromatic hydrocarbons also show limited solubility in glycols. As a result, if the feed gas contains even small amounts of aromatic compounds, non-negligible amounts of these may be absorbed. Depending on Environmental regulations, the exhaust gases from a glycol dehydration process may need to be monitored and/or treated for aromatic compounds.

Additional information on optimization of dehydration systems can be found within the ProMax Help or in the BRE 231: Sour Gas Processing course.

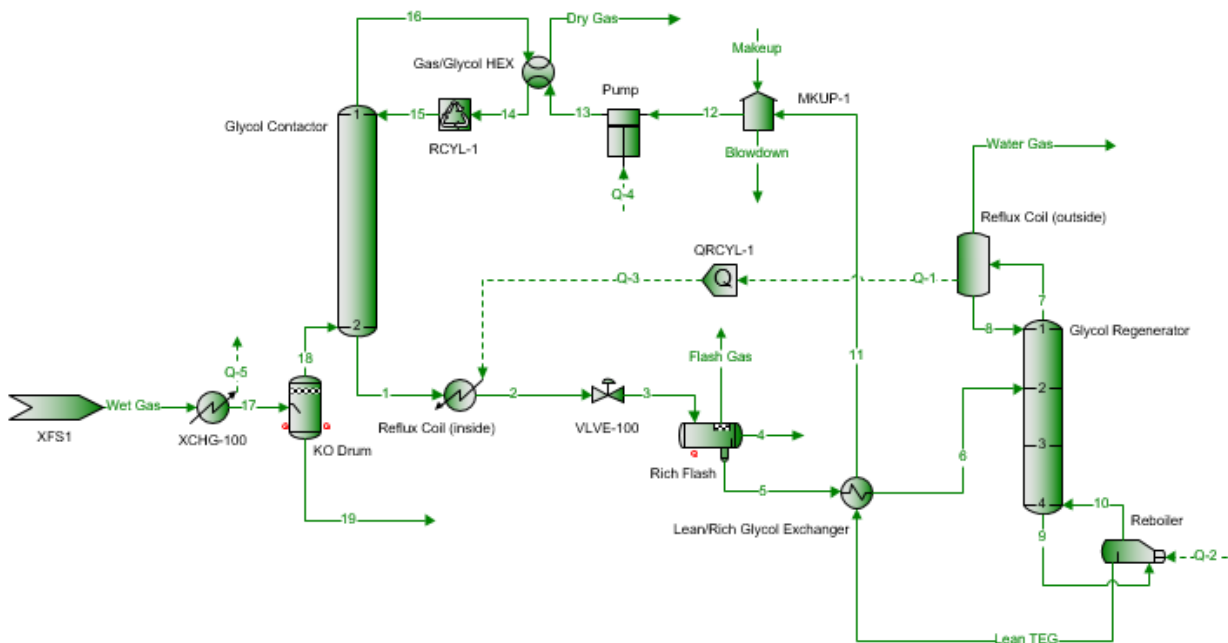
Goal and Information

Simulate a TEG unit that dries the inlet gas to a water content of less than 7 lb H₂O/MMSCF (110 mg/scm).

Part A: Ideal Stage Model

- Use the appended project created in the previous exercise. If starting here, use the conditions in the table and add a Saturator block to saturate the inlet with water.
- The Wet Gas stream is cooled to 100°F [38°C].
- Specify a complete guess for the recycle outlet.
- Use 5.6 sgpm [1.3 m³/h] for the glycol circulation rate. Makeup is 99.9 wt% TEG and 0.1 wt% H₂O.
- The absorber has 2 ideal stages, and the regenerator has 4 ideal stages.
- The rich flash operates at 60 psig [5.2 bar].
- The rich feed to the regenerator is heated to 300°F [150°C].
- The regenerator reboiler operates at 390°F [200°C].
- The regenerator operates at atmospheric pressure with a small reflux ratio generated from the reflux coil.
- Use an estimate of 20,000 BTU/h [6 kW] inside the Q-Recycle block.

*Only use table if previous exercise not worked		
Inlet Conditions		
Temperature	125°F	[50°C]
Pressure	990 psia	[68.5 bara]
Flow	96.6 MMSCFD	[107 800 Nm ³ /h]
Composition	Mole %	
H ₂ S	3.3 ppm	
CO ₂	1.31	
Methane	86.63	
Ethane	5.87	
Propane	4.23	
n-Butane	0.94	
n-Pentane	0.52	
n-Hexane	0.33	
n-Heptane	0.16	
Benzene	0.005	
Toluene	0.002	
Ethylbenzene	0.001	
o-Xylene	0.002	



QUESTIONS

1. What is the water content (lb/MMSCF) of the dry gas? The water mass concentration in the Lean TEG stream? Display these in Callout Boxes on the flowsheet.
2. At what temperature will hydrates form in the dry gas?
3. What is the mass flow of water in the feed gas?
4. How much BTEX (ton/yr) leaves in the Water Gas stream?
5. Create a customized ProMax report for the project.

Part B: Packed Tower Using Mass + Heat Transfer Model

One advantage of the Mass + Heat Transfer Model available in ProMax is the ability to directly specify a height and type of packing in the column. Change the contactor tower to use the “Mass + Heat Transfer” model, and select the “General Mass + Heat Transfer” column type.

- Increase the column “stages” to 6. This number for a packed column represents the number of calculation increments. Generally, each increment should represent approximately 18 in. [0.5 m] of packing.
- From the “General” selections, choose “Structured” packing, a 4.25 ft. [1.3m] column diameter, 10 ft. [3m] column height, and a system factor of 1 for structured packing.
- From the “Structured” selections, choose “Sulzer Mellapak® 350.X metal” as the packing type.
- Solve the flowsheet again. How do the results compare?

Optional Extended Exercise – Stripping Gas in Dehydration

Goal: See how the use of stripping gas improves the performance of this model.

PROCESS INFORMATION

- Modify the process to feed 18 SCFM [34 scm/h] of the dry gas stream as stripping gas into the reboiler (approximately 3 SCF / gallon glycol circulation). Preheat the stripping gas stream to 350°F [177°C], and then lower the pressure to 20 psig [1.4 barg]. What is the new water content of the dry gas?

FAQs on Exercise 5

- Why is there no flow from the rich flash tank to the regenerator?
This generally occurs if a 3-phase separator is used, but the current conditions only result in 1 or 2 phases. The separator block in ProMax defaults to the “Light Liquid” outlet for the Main Liquid Phase regardless of the composition. To change the behavior of the separator, change the “Main Liquid Phase” option within the separator block to “Heavy Liquid”.
- Stream water content (lb/MMSCF) and hydrate formation temperature can be found in the “Freeze Out, Hydrate, H2O Dew Point” analysis.
- Use the “Composition Subset” analysis to quickly calculate the amount of BTEX (or other groupings) in a stream.

Exercise 6: SIMPLE TURBOEXPANDER DEMETHANIZER

Process Background: Gas Processing

Raw natural gas coming from the well will contain varying amounts of ethane, propane, n-butane, isobutane, and pentanes. These compounds comprise what is called natural gas liquids (NGLs). These NGLs are removed from the methane, which is sold as “natural gas.”

The NGLs are typically removed before the natural gas is introduced to the transmission pipelines. This helps to maintain the quality of the natural gas within regulations and minimize liquids formation in the pipeline. The removed NGLs can be used as raw material for chemical plants, in enhanced oil recovery, and as a fuel source. Thus, NGL products are often more profitable than natural gas and so improved recovery is desired.

The NGLs are removed in a demethanizer column. The methane, at pipeline purity, is recovered at the top of the distillation column. Depending on the economic situation, ethane will either be sent out the top along with the methane (known as “ethane rejection”) or recovered in the bottom liquids (“ethane recovery”). All other heavier hydrocarbons are recovered in the bottom liquid stream where they go on to be fractionated into different marketable products. Demethanizers are operated at very low temperature ranges (e.g. -150°F [-100°C] at the top) and at moderate pressures around 200-300 psig [14-21 barg].

The temperatures and reboiler requirements for the demethanizer usually result in the use of a “cold box” exchanger. This is often a compact exchanger exchanging energy with multiple streams (often 4-5 different streams). These compact exchangers can achieve closer approach temperatures than shell & tube exchangers and are designed to handle much lower temperatures. ProMax has the ability to both model and rate these exchangers.

Additional information on optimization of turboexpander systems can be found in the “BRE 201: Gas Processing” course or within the ProMax Help.

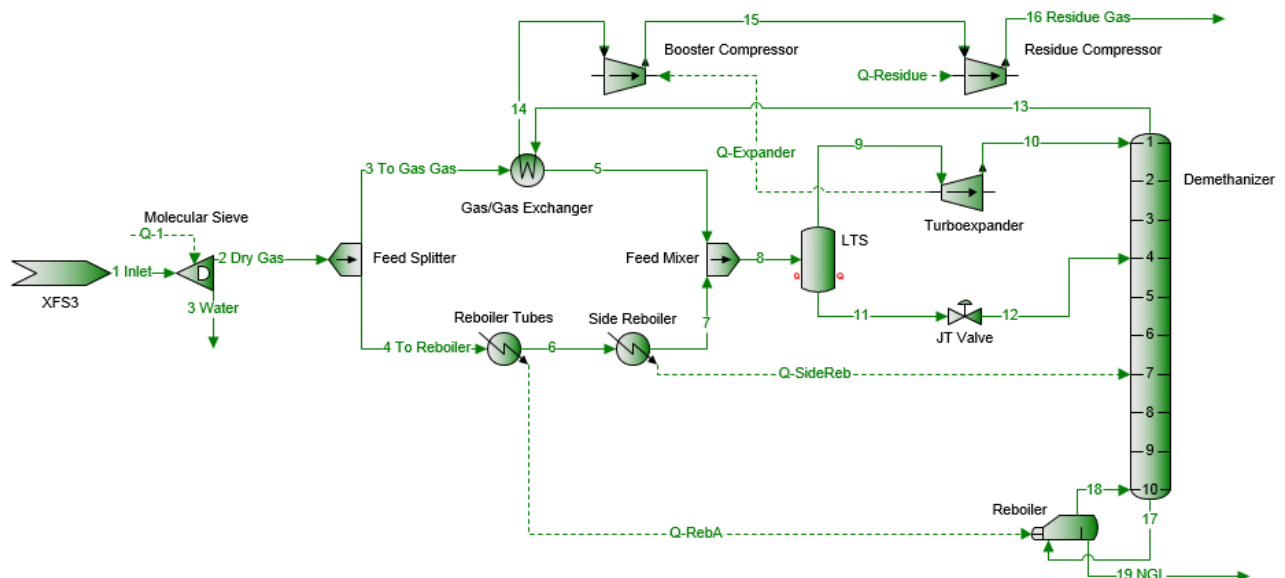
Goal and Information

This exercise continues from Exercise 5: Glycol Dehydration Unit. Alternatively, Exercise 6: may be opened and the table below may be used to specify inlet conditions.

A simple turboexpander-based demethanizer is used to recover ethane and heavier components from the gas stream as natural gas liquids (NGLs). The residue gas must be delivered at 900 psig [62 barg] with a gross ideal gas heating value of no more than 1100 BTU/ft³ [41 MJ/m³].

- Add a cross-flowsheet connector to the “Dry Gas” stream exiting the TEG unit. Connect it to a new stream on the “Turboexpander” flowsheet.
- The feed is completely dehydrated by a molecular sieve; model this with a Divider (under “Auxiliary Objects”) to remove 100% of water and TEG.
- Use a feed split of 25% to the reboilers.
- The low temperature separator (LTS) inlet stream should be set to -53°F [-47°C]. ProMax will calculate the Gas/Gas Exchanger outlet temperature required to achieve the specified LTS temperature.
- The Side Reboiler duty is 5.4 MMBTU/h [1.6 MW].
- Operate the Turboexpander at an outlet pressure of 250 psig [17.2 barg] and an 80% adiabatic efficiency.
- The JT Valve outlet pressure is 252 psig [17.3 barg].
- The bottoms liquid requirement is a C₁/C₂ ratio of 0.015, on a standard liquid volume basis.
- The compressors have a 75% polytropic efficiency. Set the residue gas pressure to 900 psig [62 barg].

*Only use table if previous exercise not worked		
Inlet Conditions		
Temperature	102°F	[39.1°C]
Pressure	965 psig	[66.5 barg]
Flow rate	96.3 MMSCFD	[107 500 Nm ³ /h]
Composition	Mol %	
H ₂ S	3 ppm	
CO ₂	1.55	
Methane	86.44	
Ethane	5.86	
Propane	4.22	
n-Butane	0.93	
n-Pentane	0.52	
n-Hexane	0.33	
n-Heptane	0.15	
Benzene	47 ppm	



QUESTIONS

1. What percentage of ethane and propane are recovered in the NGL stream? This can be found in the Recoveries grouping inside of the demethanizer block.
2. How does the booster compressor pressure change compare to the expander pressure change?
3. What are the Minimum Effective Approach Temperatures for each exchanger?
4. If a kettle-type reboiler was used instead of a plate-fin exchanger, the Reboiler would use the “Pool Boiling” heat release curve type. Change the Reboiler to “Pool Boiling” and check the new Effective End Approach Temperature. Would the current process conditions be feasible if a kettle-type reboiler was used?

Exercise 7: EXCEL IMPORT/EXPORT

Goal and Information

This exercise continues from Exercise 6: Simple Turboexpander Demethanizer. Alternatively, Exercise 7: may be opened. Use the Excel Import/Export feature to see how changing the LTS inlet temperature and turboexpander efficiency affects NGL production and required residue compression.

PROCESS INFORMATION

- Create an Excel workbook by clicking the “Add Excel Workbook” option in the ProMax Ribbon.
- Label the first two columns “LTS Temperature” and “Turboexpander Efficiency”. Below these headers, type in -53°F [-47°C] and 80%.
- Import the LTS temperature and turboexpander efficiency values from Excel. This is done by right-clicking on these properties within ProMax. In order to import values from Excel, first delete the user-defined specifications input for the LTS temperature and turboexpander efficiency inside ProMax.
- Label three more columns “Compressor Power”, “NGL Flow Rate”, and “Residue Gas Composition”. Export the residue compressor power and NGL flow rate from ProMax into these columns in Excel. Make sure the selected units are correct. Export the entire Residue Gas composition (mol%) into the last column.

LTS Temperature	Turboexpander Efficiency		Compressor Power	NGL Flow Rate	Residue Gas Composition
°F	%		hp	bbld	mol%
-53	80				

QUESTIONS

1. What happens to the residue compression power when the turboexpander efficiency drops to 70%?
2. What happens to the NGL flow rate when the LTS temperature increases to -45°F [-43°C]?

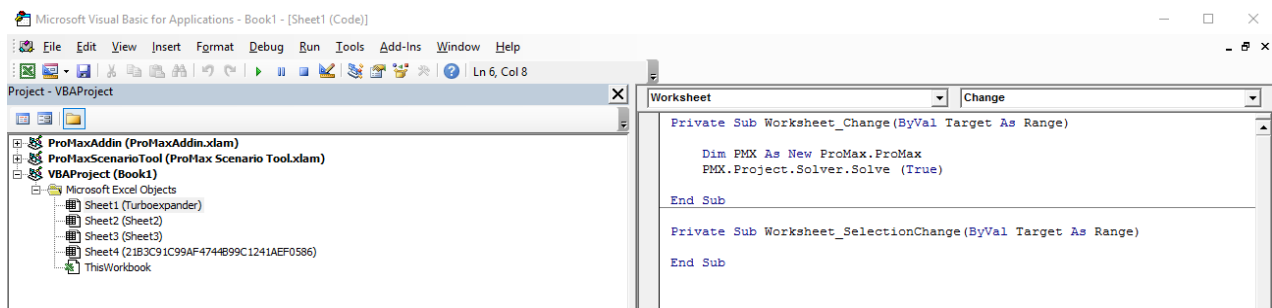
Optional Extended Exercise – Simple VBA Interaction

There are many things that some simple programming can simplify, and the Microsoft Office products provide easy access to Visual Basic for Applications.

With just 2 lines of code in the current Turboexpander exercise, ProMax will automatically execute after any change in Excel.

- In the excel workbook, type <Alt>+<F11> to open the VBA interface. This window can also be opened in the Developer tab by clicking the Visual Basic button
- In the VBA window, go to the Tools tab and select References.
- Within the available references list, select the “BR&E ProMax Type Library”.
- On the left side of the VBA window, double click on “Sheet1” to write code for this Excel sheet.
- Directly above the blank white page that appears, there are two dropdown menus. Set the left dropdown menu to “Worksheet” and then the right dropdown menu to “Change”. This creates a private sub where code is written that will go into effect whenever there is a change on the worksheet.
- Inside the “Private Sub Worksheet_Change”, write the following two lines:

```
Dim PMX As New ProMax.ProMax  
PMX.Project.Solver.Solve (True)
```



- Return to the Excel worksheet and change the LTS temperature to -55°F [-48°C]. ProMax will now immediately re-execute after this change has been made.

Exercise 8: SULFUR RECOVERY UNIT

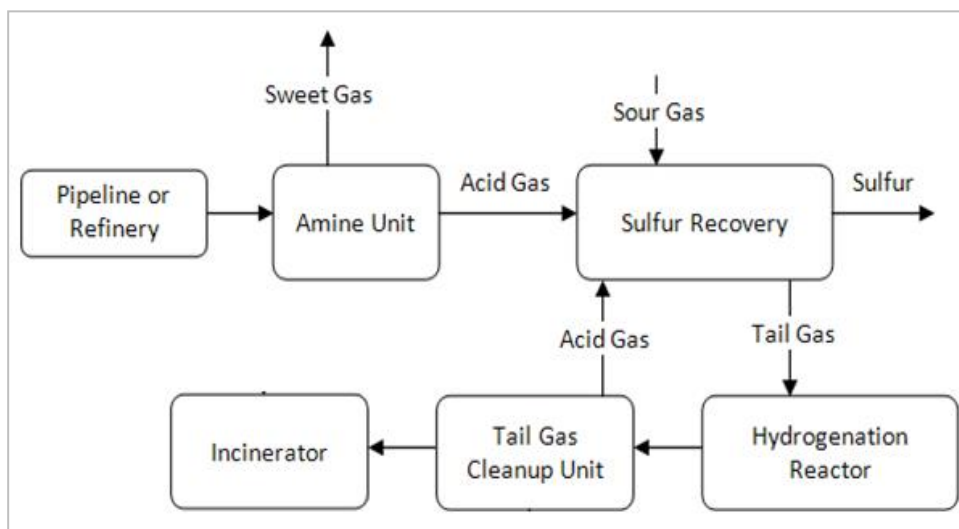
Process Background: Sulfur Recovery

If sufficient H_2S is present, refineries and gas plants will require sulfur recovery units. These units take the hydrogen sulfide captured from the process and convert it to elemental sulfur for sale and transport. Claus units are the most common sulfur recovery process. This process entails burning the acid gases in a burner, then passing the gas over a specialized catalyst that converts the sulfur species into elemental sulfur. The gas is then cooled in sulfur condensers, and the liquid sulfur is removed. A typical Claus unit repeats this process several times with several catalytic beds, taking advantage of the equilibrium shift each time sulfur is removed via a sulfur condenser.

There are several variations on the typical Claus plant, including sub-dewpoint Claus beds, direct oxidation reaction beds, and partial oxidation reaction beds, any of which can be modeled in ProMax. ProMax models these reactors using Gibbs minimization with preconfigured Gibbs sets to represent each reactor in a sulfur recovery unit.

In many cases, after the acid gases have been sent through the catalytic reactor beds, the tail gas still contains too much sulfur to incinerate. The most common method of alleviating this concern is to send the tail gas through a hydrogenation reactor, where hydrogen is added to convert most of the remaining sulfur species into hydrogen sulfide. The tail gas is then fed to an amine system for selective H_2S removal. The recovered H_2S is recycled back through the sulfur recovery unit, and the remaining off gas from this tail gas clean up unit can then be incinerated. This entire process can be modeled in ProMax by combining the various parts.

Additional information on optimization of sulfur recovery systems can be found within the ProMax Help or in the BRE 231: Sour Gas Processing course.

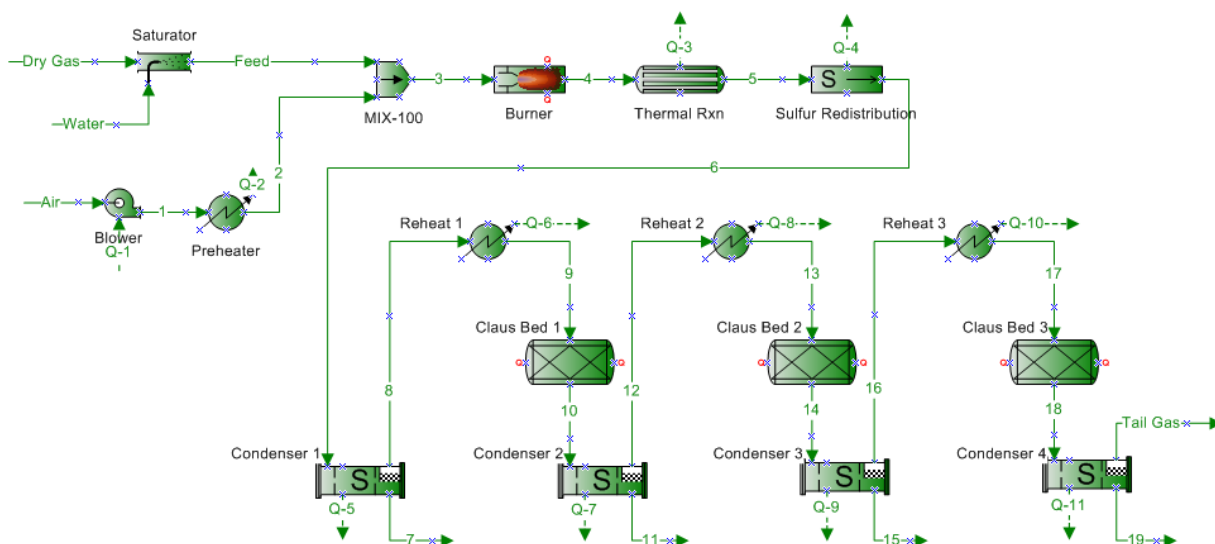


Goal and Information

This exercise simulates a three-bed Claus sulfur recovery unit. All reactors in this exercise are Gibbs Minimization reactors, utilizing specialized Gibbs Sets that are preconfigured to select the reactive species and to add the necessary constraints.

- Set the Environment to “Sulfur ASRL”
- Additional components to include in the Environment: H_2 , SO_2 , CO , COS , CS_2 , $\text{S}_1 - \text{S}_8$
- The blower increases the air pressure to 16 psig [1 barg]. The preheater then heats it to 500°F [260°C] with a $\Delta P = 0.5$ psi [0.03 bar].
- The Burner uses the “Acid Gas Burner” Gibbs Set to create COS and CS_2 reaction constraints. Assume a 0% bypass fraction and $\Delta P = 0$.
- The Thermal Reaction zone (1st pass of the waste heat boiler) produces an outlet temperature of 1200°F [650°C]. This uses the “Sulfur Thermal Reaction Zone” Gibbs Set and has a pressure drop of 0.5 psi [0.03 bar]. In this set, certain components are “quenched” to keep from reacting below specified temperatures, better reflecting actual performance. COS and CS_2 are excluded from reacting.
- The Sulfur Redistribution block (2nd pass of the waste heat boiler) produces an outlet temperature of 700°F [370°C]. This block uses the “Sulfur Redistribution” Gibbs Set. The pressure drop is 0.5 psi [0.03 bar].
- Specify the Claus Beds, reheats, and condensers sequentially, using the following information:
 - All sulfur condensers use the Gibbs Set “Sulfur Condenser” with a 0.5 psi [0.03 bar] pressure drop. Condensers 1 and 2 have an outlet temperature of 320°F [165°C], while condensers 3 and 4 have outlet temperatures of 300°F [160°C].
 - Reheats have a pressure drop of 0.5 psi [0.03 bar]. “Reheat 1” increases the temperature to 500°F [260°C], “Reheat 2” heat the stream to 480°F [250°C] and “Reheat 3” heats it to 430°F [220°C].
 - The Claus beds have a 5% bypass fraction and a 0.2 psi [0.01 bar] pressure drop. Claus Bed 1 uses the “GPSA Hydrolyzing Claus Bed” Gibbs set to partially destroy COS and CS_2 . Beds 2 and 3 use the “Claus Bed” Gibbs Set as they do not have the catalyst and temperature required for COS and CS_2 destruction.

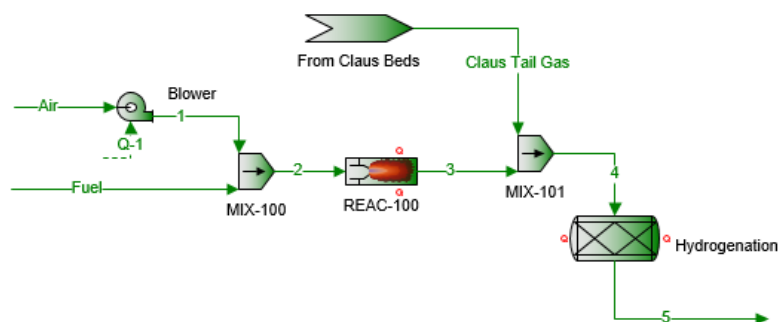
Stream	Dry Acid Gas	Air
Temperature	115°F [45°C]	77°F [25°C]
Pressure	15 psig [1 barg]	1 psig [0.1 barg]
Flow rate	165 lbmol/h [75 kmol/h]	200 lbmol/h [100 kmol/h]
Composition	Mol %	Mol %
Hydrogen Sulfide	55	0
Carbon Dioxide	43	0
Methane	1	0
Ethane	0.6	0
Propane	0.4	0
Oxygen	0	21
Nitrogen	0	78
Water	0	1



QUESTIONS

1. What is the burner outlet temperature?
2. What is the molar flow for H_2S and SO_2 in the tail gas?
3. How many lb/h of COS and CS_2 are formed in the burner?
4. How many long tons per day (tonb/d) are produced from each condenser?
5. Plot the sulfur dew point and bed temperature vs. increment for Claus Bed 2.
6. For Condenser 1, plot liquid sulfur production vs. temperature.
7. Create a recovery in the project to represent the total sulfur recovery from all 4 sulfur condensers. Display this recovery on the flowsheet by creating a short moniker for this value.

Optional Extended Exercise



PROCESS INFORMATION

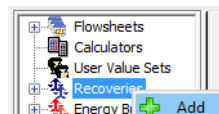
- Use a cross-flowsheet connector to send the Tail Gas on to the “Hydrogenation” flowsheet.
- The fuel is methane at 30 psig [2 bar] and 77°F [25°C]. The flow rate is 1.5 lb-mol/h [1 kmol/h].
- Ambient air, with a flow rate of 13.5 lb-mol/h [9 kmol/h], is compressed to 12 psig [1 barg] by the blower.
- The Burner uses Gibbs Minimization with the “Burner” Gibbs set. Specify a pressure drop of zero.
- The Hydrogenation Reactor is a Gibbs Minimization reactor and the Gibbs Set is “Sulfur Hydrogenation”. Set the pressure drop to 0.

QUESTIONS

1. What is the burner outlet temperature?
2. What is the remaining H_2S content in the outlet? Is all the SO_2 converted in the reactor?

FAQs on Exercise 8

- The sulfur dew point within a Claus Bed can be calculated by performing an analysis inside of the bed. This is done on the Analyses tab by adding a “Vapor Pressure, Dew, Bubble Point” analysis and selecting the Dew Point Temperature. This value can then be graphed on the Plots tab.
- Recoveries can be created from within the Project Viewer by right-clicking the “Recoveries” section and selecting “Add”. In this exercise, the reference stream is the Dry Gas while the Recovery streams are all four condenser liquid outlet streams. The “Atomic Basis” checkbox should be selected.



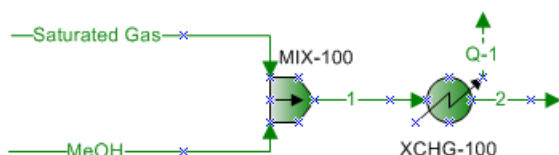
Exercise 9: SCENARIO TOOL

Part A: Methanol Injection

Use the Scenario Tool™ to study the use of methanol injection to prevent hydrate formation.

PROCESS INFORMATION

- Choose a Polar thermodynamic package.
- Inlet conditions for the Saturated Gas are given in the table to the right.
- 15 lb/h [10 kg/h] methanol is added at 80°F [27°C] and 950 psig [65.5 barg].
- The mixed stream is then cooled to 15°F [-10°C] by XCHG-100.



Inlet Conditions		
Temperature	100°F	[38°C]
Pressure	950psig	[65.5 barg]
Flow	30 MMSCFD	[35400 scm/h]
Composition	Mole %	
CO ₂	2	
Methane	85.9	
Ethane	5	
Propane	5	
n-Butane	1	
n-Pentane	0.5	
n-Hexane	0.3	
n-Heptane	0.2	
Water	0.1	

QUESTION

1. Use the Scenario Tool to vary the MeOH flowrate from 15 to 195 lb/h [10 to 100 kg/h] and compare the solid formation temperature. If the downstream pipeline requirement states that the gas must have a solid formation temperature below 0°F [-18°C], what is the optimal methanol feed rate?

Part B: Examining Amine Flowrate Effects

Finding the optimum amine flow rate for an acid gas absorber unit includes studying several variables. Having a sweet gas concentration that is within specifications is essential, but it is not the only variable to monitor. Some other variables to consider include:

- Fraction Flood – If this is an existing tower, the flooding on the trays is important to track.
- Rich Loading – especially with H₂S, if this is too high, there may be severe corrosion problems.
- Lean and Rich Approach – help determine operational stability. If either of these is near 100%, any fluctuation of the inlet feed could cause acid gas slip, resulting in a sweet gas that is no longer within specifications.

PROCESS INFORMATION

Use the Scenario Tool to track the *Sweet Gas H₂S Content*, *H₂S Lean Approach*, *H₂S Rich Approach*, *Fraction Flooding on stage 1*, and the *Total Rich Loading* for a 40 wt% MDEA solution with flow rates from 160 sgpm to 280 sgpm in steps of 10 sgpm [36 m³/h to 64 m³/h in steps of 4 m³/h]. Leave the regenerator disconnected from the system for the scenario runs.

QUESTIONS

1. At approximately what amine flow rate does the rich approach become less than 85%?
2. At approximately what amine flow rate does the absorber reach the H₂S specification of 4 ppm?
3. At approximately what amine flow rate does the rich loading reach 0.45 mol/mol?

Optional Extended Exercise: Optimum Turboexpander Pressure

A variety of factors are considered when setting the demethanizer tower pressure. Lower pressures result in colder tower temperatures, which affects the C₂ recovery while also affecting how much power is required to recompress the residue gas. Turboexpander efficiency will also affect both C₂ recovery and residue compression power.

PROCESS INFORMATION

Use the Scenario Tool to track the demethanizer overhead temperature, C₂ recovery, and residue compression power while adjusting the tower pressure (controlled by the turboexpander outlet pressure) and turboexpander efficiency. Run pressures from 225 to 275 psig [15 to 19 barg] at turboexpander efficiencies of 70% and 80%.

QUESTIONS

1. What effect does tower pressure have on C₂ recovery? On recompression power?
2. What effect does turboexpander efficiency have on C₂ recovery? On recompression power?

Optional Extended Exercise: Wellsite Heater Treater Conditions

Initial wellsite separation is performed to separate sales gas, hydrocarbon liquid, and produced water. In order to decrease the amount of flash emissions at the atmospheric storage tank, the hydrocarbon liquid is first sent to a heater treater. The vapors from the treater are compressed and re-combined with the sales gas from upstream separation. The treater pressure and temperature determine how much flash emissions will occur at the downstream hydrocarbon storage tank.

PROCESS INFORMATION

Use the scenario tool to determine what effect changing the heater treater temperature and pressure has on BTEX emissions from the atmospheric hydrocarbon tank. Run scenarios at temperatures of 125°F, 150°F, and 175°F [50°C, 65°C, and 80°C] with pressures of 20 psig, 30 psig, and 40 psig [1.5 barg, 3 barg, and 4.5 barg]. Also track the compressor power and the treater energy input.

QUESTIONS

1. Why does lowering the treater pressure decrease BTEX emissions? Is the compression power affected?
2. How does increasing the heater treater temperature affect BTEX emissions?

FAQs on Exercise 9

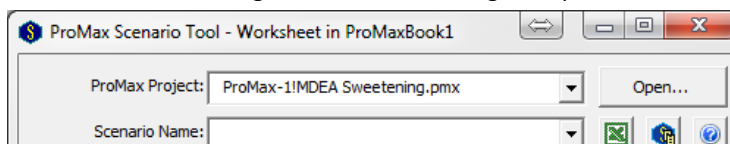
- When selecting input variables, only variables that can be user-defined within the ProMax project can be selected. These properties will be shown in **bold** lettering. Properties that are calculated by ProMax cannot be selected as Input variables and therefore are not bolded in the variable selection window.
- Multiple scenarios can be made within the same Excel workbook. To create a new blank scenario, overwrite the current Scenario Name with a new name for the new scenario. Click Enter and then select “No” from the message window that appears.

Using the Scenario Tool™ in ProMax

1. Open Excel (can be an embedded or external Excel workbook) and look for the tool under the ProMax ribbon.



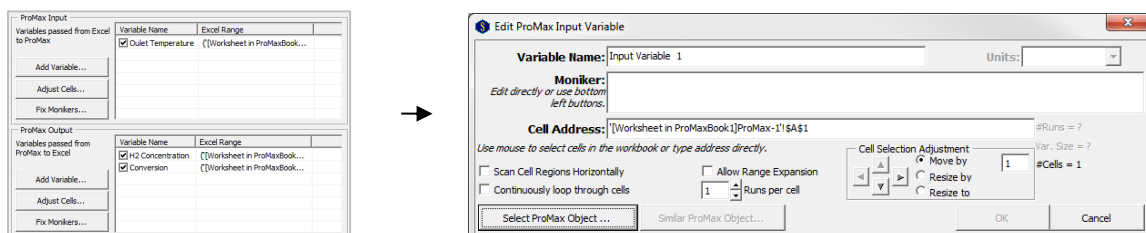
2. Start the Scenario Tool in Excel by clicking on the “Scenario Tool” option.
3. Select from the drop-down list the appropriate ProMax file to be used. A name can be given to the scenario, and the “Manage Scenarios” button to the right allows for handling multiple scenarios.



4. Organize in Excel the input parameters to be supplied to ProMax and the output locations to display the calculated results.

	ProMax Inputs		ProMax Outputs			
Run	Outlet T	Flow Rate	H ₂ S	CO ₂	Rich Loading	Reboiler
1	500	95				
2	525	100				
3	550	105				

5. Add the Input variables to the Tool by using the “Add Variable” button in the ProMax Inputs section. Use the “Select ProMax Object” button to select the desired ProMax variable. Name the variable, select the desired Excel range for this variable, and verify the units are correct.



6. Add the Output variables by using the “Add Variable” button in the ProMax Outputs section.
7. Run the tool, selecting which set of “runs” to execute.



Tip

The EZ Setup Tool opens a “wizard” to help set up a scenario step-by-step, and the 2D Grid Setup Tool is designed for easily setting a grid with two input variables and one output variable, respectively.

Exercise 10: SIMPLE SPECIFIERS

Open the unsolved Exercise 10: “Simple Specifiers” file. Simple Specifiers are utilized to set one variable based on another already known variable. This automates a project so that when one variable is changed other variables will adjust accordingly.

Part A: Lean Amine Temperature

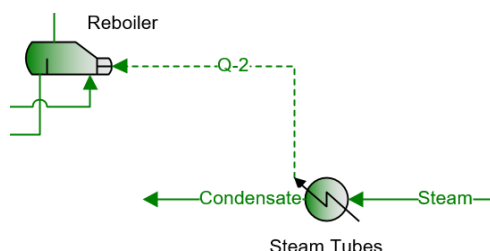
Add a Simple Specifier to maintain the lean amine contactor feed 10°F [5°C] above the sour gas feed temperature.

QUESTION

1. If the feed gas temperature decreases to 90°F [32°C], what is the new sweet gas H₂S compositions?

Part B: Reboiler Duty

Add a Simple Specifier to set the reboiler duty based on 1 pound of steam per gallon of amine solution in circulation [0.13 kg/L solution]. To do this, attach the reboiler block energy stream to a new single-sided exchanger to represent the steam tubes side. Use 50 psig [3.5 barg] saturated steam at the inlet and saturated water at the outlet.



QUESTIONS

1. What are the sweet gas CO₂ and H₂S compositions?
2. What is the mass flow of steam required?
3. What is the reboiler duty?

Optional Extended Exercise: Turboexpander JT Valve Pressure

On the second flowsheet in this exercise, there is a turboexpander unit. Add a Simple Specifier to maintain the pressure out of the JT valve 3 psi [0.2 bar] higher than the turboexpander outlet pressure.

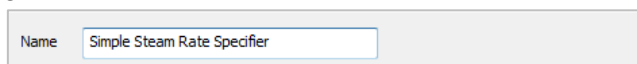
QUESTIONS

1. What happens to the ethane recovery and residue compression requirements if the turboexpander outlet pressure is dropped to 230 psig [16.5 barg]?
2. What equipment issues has this lower pressure (and thus lower temperature) potentially introduced?

Using Simple Specifiers in ProMax

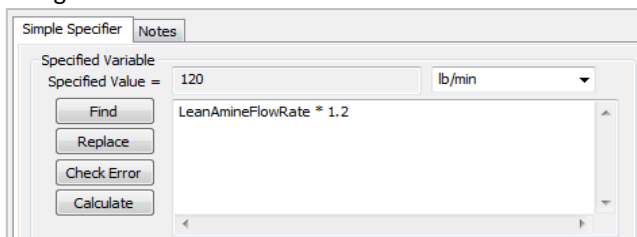
A Simple Specifier is used to set the value of a property based on the value of other known variables or constants. To set a specifier, right-click on the property to be calculated, then select *Create Simple Specifier*.

1. Name the Simple Specifier.



Simple Specifier dialog box. The Name field contains the text "Simple Steam Rate Specifier".

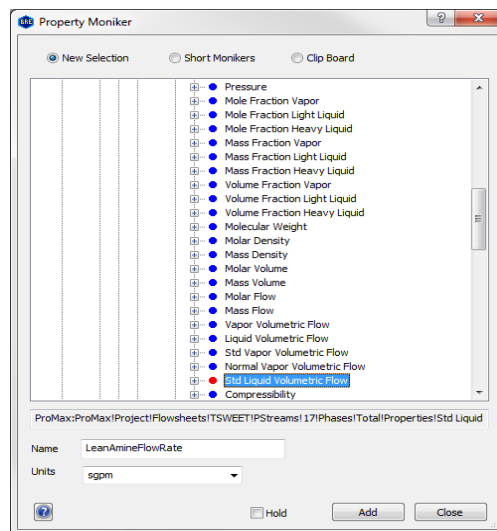
2. Write the expression that defines the specified variable. This must be a single-line expression, but can have multiple independent variables. Note that the independent variables must be known and calculated prior to the specified variable being used.



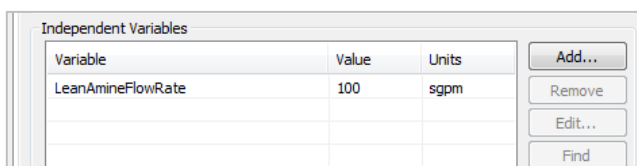
Simple Specifier dialog box. The Specified Variable field contains the expression "LeanAmineFlowRate * 1.2". The unit dropdown is set to "lb/min". Buttons for Find, Replace, Check Error, and Calculate are visible.

3. A unit selection is also made. ProMax will maintain this selection even if project units are altered.
4. Select the Add button to add the Independent Variables to be used in the specifier. The Property Moniker dialog box, as shown to the right, will appear.

5. From the variable selection dialogue, begin expanding the tree through the project, flowsheets, stream or block, down to the desired independent variable. When a valid selection is picked, the gray box below the tree will fill in with a moniker string.
6. Upon selecting the variable, type a unique name to briefly identify the value. This name must not include any special characters, with the exception of the underscore (`_`); a space will be replaced with an underscore. The name must not start with a number and is case-sensitive.
7. Select Add when finished, and repeat for as many variables as required for the specifier. When an independent variable has been selected, it will appear in the Independent Variables list with its name, value, and units as shown below.



Property Moniker dialog box. The 'New Selection' tab is active. A list of variables is shown, including Pressure, Mole Fraction Vapor, Mole Fraction Light Liquid, Mole Fraction Heavy Liquid, Mass Fraction Vapor, Mass Fraction Light Liquid, Mass Fraction Heavy Liquid, Volume Fraction Vapor, Volume Fraction Light Liquid, Volume Fraction Heavy Liquid, Molecular Weight, Molar Density, Mass Density, Molar Volume, Mass Volume, Molar Flow, Mass Flow, Vapor Volumetric Flow, Liquid Volumetric Flow, Std Vapor Volumetric Flow, Normal Vapor Volumetric Flow, Std Liquid Volumetric Flow, and Compressibility. The 'Add' button is visible at the bottom right.



Independent Variables dialog box. The table contains the following data:

Variable	Value	Units
LeanAmineFlowRate	100	sgpm

Buttons: Add..., Remove, Edit..., Find



Tip

Selecting the "Hold" check box in the Property Moniker dialog box will keep the dialog open after adding the current variable, simplifying the process of adding multiple independent variables.

Exercise 11: SIMPLE SOLVERS

Simple Solvers adjust one variable in order to achieve a target value for another variable in a project. The user must set an initial guess so that the Simple Solver can iterate until a final solution is reached. This exercise demonstrates the use of Simple Solvers in a variety of processes previously discussed in this course.

Part A: Amine Rich Loading Solver

Open the unsolved Exercise 11: “Simple Solvers” file. Add a Simple Solver to calculate the amine circulation rate needed to attain a rich amine loading equal to 0.40 mole acid gas/mole amine. Remember to assign a priority for this solver.

QUESTIONS

1. Why would a Simple Specifier not work in this situation? Why is a solver necessary?
2. What is the amine circulation rate to achieve a 0.40 rich loading?
3. What are the sweet gas CO₂ and H₂S compositions?

Part B: Amine Lean Loading Solver

Delete the Simple Specifier on the steam rate. Create a Simple Solver to determine the required mass flow of steam to achieve a lean loading of 0.0035 mole/mole total acid gas loading. Please review the priorities required for each solver for efficient execution of the project.

QUESTIONS

1. What are the sweet gas CO₂ and H₂S compositions when using both solvers?
2. What is the mass flow of steam required to attain the new lean loading?
3. What is the reboiler duty?

Optional Extended Exercise: SRU Tail Gas Ratio Solver

On the second flowsheet, add a Simple Solver to automatically adjust the air flow rate to attain an H₂S/SO₂ tail gas ratio of two. Make note of the original air flow rate and overall sulfur recovery prior to creating this solver.

QUESTIONS

1. What is the required air flow rate? What percentage increase of air flow was required?
2. What is the new overall sulfur recovery?

Optional Extended Exercise: Turboexpander Exchanger Approaches

On the Turboexpander flowsheet, add Simple Solvers to maximize heat integration in the turboexpander plant. Control the LTS inlet temperature such that the minimum effective approach temperature of the gas/gas exchanger is 5°F [3°C]. Set a similar solver controlling the side reboiler duty to achieve a minimum effective approach temperature of 5°F [3°C]. A third solver is set on the inlet split percentage to the reboilers to achieve a minimum effective approach temperature in the bottom reboiler of 5°F [3°C]. Set the priority for each solver to zero.

QUESTIONS

1. Why is the priority set to be the same for each solver?
2. With these solvers in place, increase the turboexpander outlet pressure to 275 psig. Note how this affects the LTS temperature and the ethane recovery.

Optional Extended Exercise: Pipeline Network Delivery Pressure

On the Pipeline flowsheet, use a specifier to make the “Gas 3” stream pressure equal to stream 2 and another specifier to make the “Gas 2” stream pressure equal to the “Gas 1” stream pressure. Put a Simple Solver on the “Gas 1” stream pressure to get a final delivery pressure of 500 psig [34.5 barg] to the facility.

QUESTIONS

1. Why are the Simple Specifiers necessary in this situation?
2. What is the required “Gas 1” stream pressure to achieve the desired delivery pressure?

Optional Extended Exercise: JT Skid – Methanol Injection

As discussed in Exercise 9: Scenario Tool, methanol can be injected into a process stream to prevent hydrate formation when necessary. On the JT Skid flowsheet, place a solver on the methanol mass flow rate to determine the required methanol injection rate so that stream 5 will be 10°F above solid formation. Use a step size of 10 lb/h [5 kg/h].

QUESTIONS

1. How much methanol needs to be injected?
2. OPTIONAL: Put solvers on the Gas/Gas and Gas/Liquid exchanger outlet temperatures so that both exchangers have approach temperatures of 10°F [5°C].

Using Simple Solvers in ProMax

A Simple Solver is used to adjust one variable in order to hit a goal or spec on another variable in a process. A Simple Solver must be used if ProMax will need to iterate to find the desired solution.

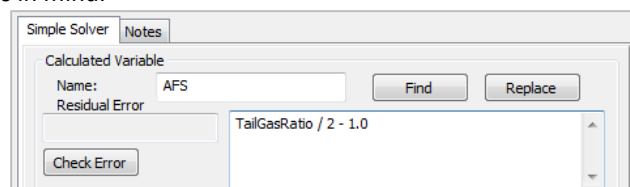
For example, if the goal is to adjust the flow rate of air fed to a Claus unit to achieve a set tail gas ratio of H₂S to SO₂, a Simple Solver could be placed in the inlet air flow rate field. ProMax will use the solver criteria to iterate on the air flow rate until it achieves the desired tail gas ratio.

To create a Simple Solver:

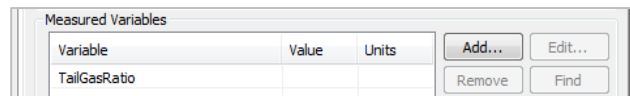
1. Right-click on the property that the solver is to adjust and select *Create Simple Solver*.
2. Name the Simple Solver.



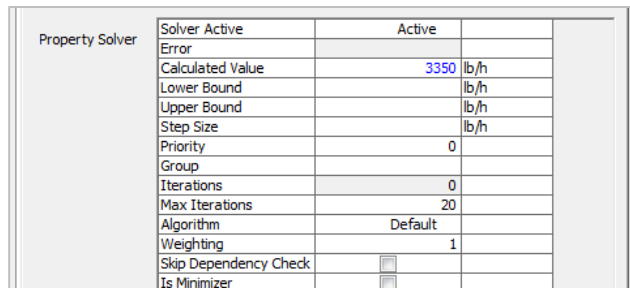
3. ProMax will iterate the Calculated Variable until the equation written here is equal to zero, so the equation must be written with this in mind.



4. Select the Add button to open a tree dialog as shown in the Specifier example on page 33. Add the Measured Variable to be used. This is the value ProMax will measure to see if it meets the goal (in this case, the tail gas ratio).



5. A guess for the “Calculated Value” must also be supplied to give the solver a starting point. Upper and lower bounds may be set, though this is usually not necessary and will cause the solver to fail if the solution is outside of the given bounds.



6. A Priority must be set if multiple solvers are present in the simulation for acceptable execution time. It is suggested that the solvers be prioritized to solve sequentially, with loops solving from the “inside out”.

Writing the Simple Solver Expression:

1. For this example, the Tail Gas Ratio should equal 2 when solved: $TailGasRatio = 2$
2. Since ProMax expects to solve at zero, this is rewritten: $TailGasRatio - 2 = 0$
3. ProMax needs only the function written, so the “0 = ” is omitted: $TailGasRatio - 2$



Tip

While writing the function as “*TailGasRatio - 2*” is acceptable, it is highly recommended that each Simple Solver expression be “normalized” to aid in convergence. Normalizing the above expression results in the following: “*TailGasRatio/2 - 1*”

Exercise 12: USER VALUE SETS

A new user-defined variable can be specified to represent any desired value that is not defined within ProMax by default. The following exercise shows some examples of how this can be useful.

Part A: Steam Ratio

Add a user-defined variable to represent a reboiler steam ratio (pounds of steam per gallon amine in circulation). Set this ratio to 1.2 lb/gal [0.15 kg/liter]. Put a Simple Specifier on the reboiler steam rate on the MDEA flowsheet to utilize this ratio.

QUESTIONS

1. Use a “property input” stencil to display the steam ratio on the flowsheet. Adjust the ratio to 1.3 lb/gal [0.163 kg/liter] and note the change to the lean loading and Sweet Gas H₂S concentration.

Part B: Cost Estimation

Some operations can estimate an operating cost based on the BTUs required. Create a user-defined variable to input this energy cost; start with a cost of \$6.00/MMBtu. Add an additional user-defined variable in the same set to display the cost of running the amine reboiler based on this information.

QUESTIONS

1. What is the current operating cost of the amine reboiler (\$/day), based only on the required energy? What would the operating cost be if the energy cost was \$6.50/MMBtu?

Optional Extended Exercise: Refrigeration Loop Compression Power

Create a user-defined variable to sum the total compressor horsepower requirement for the Refrigeration flowsheet. Once created, add the value as a short moniker in the project. Then add a property table to display this value on the flowsheet, based on the short moniker name.

QUESTIONS

4. What happens if an upper bound of 225 hp [168 kW] is set?

Optional Extended Exercise: Glycol Flow Rate Ratio

Glycol flow rates are typically set according to the amount of water in the feed gas. Typical ratios are between 2 and 5 gallons of glycol for each lb of water in the feed gas [15-40 L/kg]. On the dehydration flowsheet, add a user-defined variable to represent a glycol flow ratio of 2 gal/lb [15 L/kg]. Put a Simple Specifier on the glycol flow rate to utilize this ratio.

QUESTIONS

5. How are the BTEX emissions affected when this ratio is changed to 3 gal/lb [24 L/kg]?

Exercise 13: INCINERATOR

Process Background: Incineration

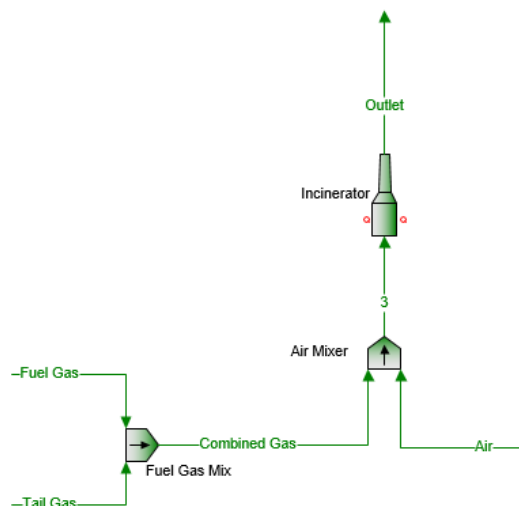
As discussed previously, sulfur recovery units convert hydrogen sulfide captured from upstream processes and convert it to elemental sulfur for sale and transport. The tail gas, with small amounts of hydrogen sulfide remaining after processing, is typically sent to an incinerator. Tail gas generally has a very low heating value due to large concentrations of incombustibles so fuel is often mixed with the tail gas in order to ensure combustion. Sufficient air must also be supplied in order for enough oxygen to be present for complete combustion of hydrogen sulfide.

Goal and Information

This exercise shows how to model an incinerator to combust tail gas from a sulfur recovery unit. The effects of fuel gas and air flow rates will be analyzed and discussed.

PROCESS INFORMATION

- The tail gas stream is pre-configured to represent tail gas leaving a sulfur recovery unit.
- Temperature, pressure, and composition are pre-defined in the air and fuel gas streams.
- To begin the exercise, set the fuel gas flow rate to 0 lbmol/h [kmol/h].
- Use a Simple Specifier on the air flow rate to mix 20% excess oxygen with the combined gas. The amount of oxygen necessary for combustion can be found by performing a Combustion analysis in the combined gas stream.
- The incinerator is modeled as a Gibbs Minimization reactor, using the Burner reaction set.
- Set a 0 psi pressure drop in the incinerator. Set the bypass fraction to 2%.



QUESTIONS

1. What is the current incinerator outlet temperature? Why is a higher temperature desirable?
2. Fuel gas is used to achieve a sufficiently high temperature to ensure combustion. Put a Simple Solver on the fuel gas flow rate to get an incinerator outlet temperature of 1600°F [870°C]. What is the calculated fuel gas flow rate to achieve an outlet temperature of 1600°F [870°C]?
3. Change the air flow rate to run at 50% excess oxygen and re-converge the simulation. How does the required fuel gas flow rate change?
4. What does a bypass fraction of 2% represent?
5. What are the SO₂ emissions from the incinerator in ton/yr?

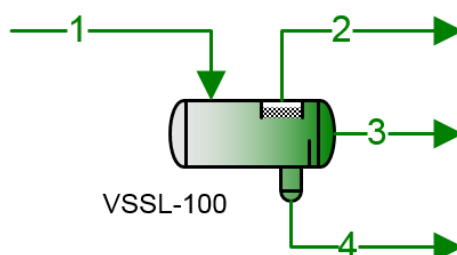
Exercise 14: SEPARATOR SIZING

Use the following information to size a 3-phase horizontal bucket & weir separator.

PROCESS INFORMATION

- Create a hypothetical component to represent the C6+ with a molecular weight of 180 lb/lbmol and API gravity of 42.
- The feed conditions and composition are shown in the attached table.
- Check the “Include Separator Sizing” box inside the separator on the Process Data tab.
- On the Sizing tab, select “Horizontal 3-phase Bucket & Weir” for the Vessel Type
- Use a K-value of 0.5 ft/s [0.15 m/s] for V-L separation.
- Use a liquid droplet diameter of 150 micron for L-L separation.
- Both the light liquid and heavy liquid holdup times are 0.5 minutes and surge times are 0.25 minutes.
- The light liquid low level shutdown depth is 6 inches [15 cm].
- Demister pad thickness & overhead clearance are 6 inches [15 cm].
- Light and heavy liquid residence times are 3 minutes.
- Design temperature is 200°F [93°C] and MAWP is 435 psig [30 barg].
- Corrosion allowance is 0.0625 inches [0.15 cm].
- Use carbon steel A516-70 for the shell and head material of construction.
- Use ellipsoidal heads.

Inlet Conditions		
Temperature	155°F	[68°C]
Pressure	415 psia	[29 bar]
Flow Rate	58,225 lbmol/h	[26 410 kmol/h]
Composition	Mole %	
Methane	4.9	
Ethane	1.9	
Propane	1.5	
n-Butane	1.0	
n-Pentane	0.5	
Water	84.5	
C6+	5.7	



QUESTIONS

1. What is the calculated L/D ratio?
2. In the range $1.5 < L/D < 6$, what is the smallest (by mass) separator that will meet design requirements? What is the mass when the vessel is filled with water?
3. What are the shell and head thicknesses?
4. What is the settling compartment length?
5. Create an entrainment within the separator to make the heavy liquid outlet stream contain 1 vol% light liquid. Set this entrainment up on a volume per volume basis with 0.01 gal light liquid/0.99 gal heavy liquid.
6. Does this entrainment affect the sizing results?

Exercise 15: DEPRESSURIZATION

In this exercise, the blowdown orifice for depressurizing a vessel will be sized to a target pressure within a certain timeframe in case of fire using API 521.

PROCESS INFORMATION

- The drawn process stream represents the contents of the vessel. Operating conditions are pre-defined as described in the table.
- Drag the “Depressurization Example” shape from the ProMax Property Stencil onto the flowsheet. Select the “Vessel Contents” stream from the dropdown menu.
- The vent phase should be set to vapor.
- Set the vessel volume to 400 ft³ [11.33 m³].
- The initial pressure and temperature are the pressure and temperature of the vessel when relieving begins. Note that this is not the normal operating pressure and temperature. For this case, set the vessel relief pressure at 688 psig [48 barg] with a relief temperature of 138°F [59°C].
- Pipe inner diameter is 2 inches [6 cm]. Vent Pressure is atmospheric.
- Target Pressure will be half of the relief pressure (344 psig [24 barg]). Set the target time to 15 minutes.
- Check the “Use API521 for heat estimation?” box. Tank shape is “Horizontal Cylinder”. Fire-fighting is available.
- Tank surroundings is set to bare. Tank elevation is 2 ft [0.6m] above grade. Specify the tank length to radius ratio as 8. Check the “Use EOS for liquid phase” selection.”
- Click “Solve” at the bottom of the Depressurization window to execute the stencil.

Vessel Contents	
Temperature	100°F [38°C]
Pressure	560 psig [38.6 barg]
Composition	Mole %
C1	26
C2	30
C3	15
iC4	10
nC4	15
C5	4

QUESTIONS

1. What is the initial liquid volume fraction in the vessel?
2. What is the calculated orifice diameter?
3. What is the maximum venting mass flow rate?

Optional Extended Exercise – Relief Valve Sizing

PROCESS INFORMATION

- On the Relief Valve Sizing Flowsheet, relief conditions for a C4/C5 mixture are specified in the stream according to the table.
- Add a “Relief Valve Sizing” analysis to the C4/C5 mix stream.
- Use the API520 standard and a Conventional valve.
- Relief Temperature and Set Pressure are set at stream conditions
- Fraction Over Pressure is 10%. Back pressure is atmospheric.
- The required mass flow rate is 50,000 lb/hr [22,700 kg/h].

C4/C5 Mix	
Temperature	170°F [76.7°C]
Pressure	75 psig [5.2 barg]
Flow Rate	50,000 lb/hr [22 700 kg/h]
nC4	70 mol%
nC5	30 mol%

QUESTIONS

1. What are the calculated Effective Discharge Area and Mass Flux?
2. What is the Relieving Fluid Type? Why?
3. Set Back Pressure to 20 psig [1.4 barg] and switch to Balanced Bellows. What is the new Effective Discharge Area? What is the Back Pressure Correction Factor?

Exercise 16: SHELL & TUBE HEAT EXCHANGER RATING

Use the shell & tube exchanger specification sheet provided to rate the Lean/Rich Exchanger in an amine unit.

QUESTIONS

1. What is the area available in the exchanger for heat transfer?
2. Display the percent over-design on the flowsheet.

Optional Extended Exercise: Fraction Over Design Solver

Add a Simple Solver to the shell & tube outlet temperature going into the regenerator. Set this solver to achieve a fraction over design of 10%. Remember to assign the best priority to the solver for the most efficient execution.

QUESTIONS

1. What is the new temperature going into the regenerator?

Optional Extended Exercise: Air Cooler Rating

Rate the Air Cooler (fin fan exchanger) used to cool the lean amine before entering the amine contactor.

- The Air Cooler datasheet is found after the Shell & Tube exchanger datasheet.
- This will require creating a second side to the exchanger by connecting a second single-sided exchanger to the first via a common energy stream.
- This second side represents the air side of the exchanger. Note that “air” is not a component and so its constituent components –at a minimum oxygen and oxygen – must be used.

QUESTIONS

1. What is the calculated pressure drop for the process stream?

FAQs on Exercise 16

- Tube service is set to Demand for the shell & tube heat exchanger since the rich amine stream (2) that is being heated runs through the tubes. This is due to the corrosive nature of the rich amine.
- Fouling resistance can be user-specified for side A and side B of the exchanger. Go to the Connections tab to determine which fluid is on side A and which is on side B.

Shell & Tube Exchanger Specification Sheet									
Customer:			Job Number:			Item #: Lean/Rich Exchanger			
Plant Name:			Date:			Revision:			
Plant Location:			Prepared by:			Case:			
Service:			Checked by:						
Manufactured by:			Page: 1			of		3	
1	Size: 30 - 216 (in)		Type: NEN		Connected in: 2 in Series x 1 in Parallel				
2	Surface Per Unit (ft²): 5222.90		Shells Per Unit: 2		Surface Per Shell (ft²): 2611.45				
Performance of One Unit									
4	Fluid Allocation		Shell Side			Tube Side			
5	Fluid Name:								
6									
7	Fluid Quantity, Total (lb/h):		142927			149657			
8	Vapor (In / Out) (lb/h):		0			0			
9	Liquid (In / Out) (lb/h):		142927			149657			
10	Steam (In / Out) (lb/h):		0			0			
11	Water (In / Out) (lb/h):		85598.4			85806.0			
12	Noncondensables (In / Out) (lb/h):								
13	Fluid Vaporized or Condensed (lb/h):		N/A			50.4433			
14	Steam Condensed (lb/h):		N/A			N/A			
15	Density (Liquid / Vapor) (lb/ft³):		60.1938			65.4605			
16	Viscosity (Liquid / Vapor) (cP):		0.515857			2.13460			
17	Molecular Weight, Vapor:		18.1340			19.2864			
18	Molecular Weight, Noncondensables:								
19	Specific Heat (Liquid / Vapor) (Btu/lb °F):		1.03115			0.869604			
20	Thermal Cond. (Liquid / Vapor) (Btu/h ft² °F):		0.229784			0.217943			
21	Latent Heat (Btu/lb @ °F):		At 257.3 °F			At 120.3 °F			
22	Temperature (In / Out) (°F):		257.3			120.3			
23	Pressure (In / Out) (psia):		30.6959			100			
24	Pressure Drop (Allow / Calc) (psi):		5			5			
25	Fouling Resistance (Min) ((h ft² °F)/Btu):		0.001			0.002			
26	Heat Exchanged (Btu/h):		1.17592E+07			Mtd (Corrected) (°F): 45.6830			
27	Transfer Rate, Service (Btu/h ft² °F):		49.2847			Clean (Btu/h ft² °F): 66.2425			
Construction of One Unit									
28	Design Pressure (psig)								
29	Test Pressure (psig):								
30	Design Temperature (°F):								
31	Number of Passes Per Shell:		1			2			
32	Corrosion Allowance (inches)		0			0			
33	Connections (Size / Rating):		In 10in			14in			
34			Out 10in			14in			
35	Tube No.: 760		OD (in.): 0.75		Thk (in.): 0.065		Length (ft): 18		Pitch (in.): 0.938
36	Tube Type: Plain						Material: Carbon Steel(A214)w/ld. Tube(K01807)G24, T2, W6(BPV		Layout: 30 Triangle
37	Shell:		ID (in.): 30				Shell Cover:		
38	Channel or Bonnet:						Channel Cover:		
39	Tubesheet-Stationary:						Tubesheet-Floating:		
40	Floating Head Cover:						Impingement Plate: TEMA Required		
41	Baffles-Cross: 13		Type: Segmental		½Cut (Diam/Area): 34		Spacing (in.): 15		
42	Baffles-Long: 0		Type:						
43	Supports-Tube:		U-Bend: NO				Type:		
44	Bypass Seal Arrangement:						Tube-Tubesheet Joint:		
45	Gaskets-Shell Side:						Tube Side:		
46	Gaskets-Floating Head:								
47	Expansion Joint:						Type:		
48	Code Requirements:						Tema Class:		
49	Weight Per Shell Empty:				Operating:		Full of Water:		

Specification Sheet Air Cooled Heat Exchanger									
Client					Service			Item No.	
Location					Manufacturer			Inq. No.	
Unit Air Cooler					Job No.			P.O. No.	
1	Bay Size 12 ft x 4.125 ft				Type Forced		No. Bays 1		
2	Surface Area per Unit: Finned Tube 13999 ft^2				Bare Tube 672.301 ft^2				
3	Duty 1.05473E+07 Btu/h				MTD Eff. (Cross)			65.9639 °F	
4	Transfer Rate Btu/(h*ft^2*°F)		Finned	13.5238	Bare	281.599	Service	11.6175	Clean 13.5238
5	PERFORMANCE DATA - TUBE SIDE								
6	Fluid Name (inlet stream name):			Physical Properties of Fluid		Inlet		Outlet	
7	11					Liquid	Vapor	Liquid	Vapor
8	Total Flow 122057 lb/h			Mass Flow lb/h		122057		122057	
9	Vapor Condensed lb/h			Temperature °F		200.142		110	
10	Noncondensables lb/h			Density lb/ft^3		62.0655		64.1534	
11	Inlet Pressure 1000 psig			Viscosity cP		0.83628		2.47583	
12	Spec. Pressure Drop 5 psi			Mol. Wt. lb/lbmol		27.2869		27.2869	
13	Calc. Pressure Drop 6.56043 psi			Spec. Heat Btu/(lb**°F)		0.98716		0.931161	
14	Average h 492.838 Btu/(h*ft^2*°F)			Th. Cond. Btu/(h*ft**°F)		0.233074		0.231879	
15	Fouling Resistance 0 (h*ft^2*°F)/Btu			Bubble Point °F				Pour Point °F	
16	Average Latent Heat Btu/lb			Lethal Service		Yes / No			
17	PERFORMANCE DATA - AIR SIDE								
18	Mass Flow 1.43928E+06 lb/h			Temperature In/Out (°F)		70 / 100		Pressure / Altitude 3 psig /	
19	Average h 98.2972 Btu/(h*ft^2*°F)			Fan Mass Velocity 38178 lb/(ft^2*h)				Static Pressure Loss 70.8552 in H2O	
20	Flow per Fan 479758 lb/h			Fan Face Velocity 118.423 ft/s				Velocity Press. Loss 3.75155 in H2O	
21	DESIGN - MATERIALS - CONSTRUCTION								
22	Design Pressure:			Test Pressure:			Design Temperature:		
23	TUBE BUNDLE			HEADER			TUBE		
24	Size: 12 ft x 4.125 ft x 1.6238 ft			Material:			Material: Carbon Steel A214 Wld. Tube K01807 G24, T2, W6 BPVC		
25	No./Bay: 1			Type:			Type: Straight OD: 1 in		
26	No. Tube Rows: 10			No. Passes: 6			Corr. Allow.(In/Out) in 0 / 0		
27	Arrangement	Parallel	Series	Slope wrt Horizontal: 0 °			Min. Thick. in/BWG: 0.12 / 11		
28	Bundles 1			Plug Mat.:			No./Bundle 214 Length 12 ft		
29	Bays 1			Gasket Mat.:			Pitch 2.25 in 30 Triangle		
30	Bundle Frame:			Corrosion Allow:			FIN		
31	MISCELLANEOUS			In Nozzle No./Diam.: 1 / 19.4856 in			Material: Aluminum 1100 B209 Plate, sheet A91100 O T4 BPVC		
32	Struct. Mount. Grade / Piperack			Out Nozzle No./Diam.: 1 / 19.4856 in			OD: 2.25 in		
33	Surface Prep:			Special Nozzles:			Stock Thk. 0.049 in		
34	Louvers: Automatic / Manual			Rating and Facing:			No./in 10 1/in		
35	Vibration Switches: Yes / No			TI:			Fin Design Temp.:		
36	Bug Screen: Yes / No			PI:			Code: Stamp: Yes / No		
37	Hail Screen: Yes / No			Chem. Cleaning:			SPECS.		
38	MECHANICAL EQUIPMENT								
39	FAN			DRIVER			SPEED REDUCER		
40	Mfr.:	Model		Type:			Type:		
41	No./Bay: 3	rpm:		Mfr:			Mfr:		
42	Diameter: 4 ft	No. Blades:		Model:			Model:		
43	Pitch Man. Adj. / Auto			No./Bay:			No./Bay:		
44	Angle:			Power/Driver:			AGMA Rating:		
45	Blade Material:			rpm:			Ratio:		
46	Hub Material:			Enclosure:			Support: Structure / Pedestal		
47	Power/Fan, Design: 1615.64 hp			Volt/Phase/Cycle: / /					
48	Power/Fan, Min. Amb.:								
49	Control Action on Air Failure:			Fan Pitch Min / Max / Lockup			Louvers Open / Close / Lockup		
50	Degree Control of Outlet Process Temp. (Max. Cooling):								
51	Recirculation: None / Internal / Ext. Over Side / Ext. Over End						Steam Coil: Yes / No		
52	Plot Area:						Proposal Drawing No.		
53	Bundle Weight:						Shipping Wt:		
54	Remarks:								
55									
56									
57									
58									

ProMax Additional Exercises

Exercise 17: REFRIGERATION LOOP

The Propagation Terminal block is used within ProMax when modelling closed loop systems. In this exercise, use the Propagation Terminal to determine the required refrigerant flow.

PROCESS INFORMATION

- Specify 100% propane as the refrigerant.
- Compress to 260 psia {18 bar}, then cool to 120°F {48.5°C}.
- Expand to 35 psia {2.4 bar} across the valve.
- Use 10 MMBtu/h [2.93 MW] for the Q-Load stream in this example.
- Specify the stream leaving the chiller as 100% vapor.
- Within the Propagation Terminal, check Temperature and Pressure as Propagated Variables.

QUESTIONS

1. What is the calculated flow rate of propane?
2. What is the required compressor power for this system?

Exercise 18: MERCAPTAN REMOVAL FROM LPG USING NaOH

Caustic solution can be used to remove mercaptans from a liquid or vapor feed. In this exercise, simulate mercaptan removal from an LPG stream using a caustic solution.

PROCESS INFORMATION

- Choose a proper thermodynamic package for caustic treating.
- The caustic solution should be pumped to column pressure and then cooled to 80°F [27°C].
- Specify a two-stage LLE column.
- Note that the regeneration section is ignored for simplicity on this exercise; examples of caustic regeneration can be found in the ProMax Example files.

	Sour Liquid	Lean Caustic
Temperature	80°F [27°C]	125°F [52°C]
Pressure	300 psia [20.7 bar]	30 psia [2 bar]
Flow Rate	33 sgpm [125 lpm]	1 sgpm [3.8 lpm]
Composition	Mole %	Wt %
Propane	99.973	0.00
Methyl Mercaptan	270 ppm	0.01
NaOH	0.00	10.00
Water	0.00	89.99

QUESTIONS

1. What percentage of methyl mercaptan is removed by the caustic? (Check the column recoveries)
2. What is the liquid viscosity of the lean caustic solution feeding the column?

Exercise 19: FRACTIONATION TRAIN

Fractionate the Y-Grade liquids from a demethanizer into ethane, propane, butanes, and a C₅+ condensate.

PROCESS INFORMATION

- Deethanizer
 - a. The column operates at 300 psia [20.7 bar] with a $\Delta P=7$ psi [0.5 bar].
 - b. Ethane product exits at a 65°F [18.3°C] condenser temperature.
 - c. Specify a 0.04 C₂/C₃ ratio (standard liquid volume basis) in the bottoms product.
- Depropanizer
 - a. The exchanger cools the feed to 180°F [82°C].
 - b. The column operates at 205 psia [14.1 bar] with a $\Delta P= 3$ psi [0.25 bar].
 - c. Use a reflux ratio of 2.
 - d. 99.9% of the C₃ is to be recovered in the vapor.
- Debutanizer
 - a. The column pressure is 150 psia [10.3 bar] with a $\Delta P= 5$ psi [0.4 bar].
 - b. The condenser is sub-cooled to 160°F [71°C].
 - c. Recover 99.9 mol% of the C₄ in the overhead product.
 - d. Use a boil-up ratio of 4.
- Butane Splitter
 - a. The column operates at 135 psia [9.3 bar] with a $\Delta P= 20$ psi [1.5 bar].
 - b. The i-C₄ condenser is sub-cooled to 70°F [21°C].
 - c. Use a boil-up ratio of 11.5.
 - d. The bottoms product specification is 97.5 mol% n-C₄.

Inlet Conditions	
Temperature	80°F [27°C]
Pressure	500 psia [34.5 bar]
Flow Rate	2400 lb-mol/h [1080 kmol/h]
Composition	Mole %
Methane	0.63
Ethane	49.00
Propane	27.00
i-Butane	4.15
n-Butane	7.60
i-Pentane	2.20
n-Pentane	2.14
n-Hexane	3.80
n-Heptane	0.60
Benzene	0.12
Toluene	0.17
Ethylbenzene	0.009
p-Xylene	0.13
Nitrogen	0.051
CO ₂	2.40

QUESTIONS

1. What is the ethane recovery in the deethanizer?
2. What is the propane recovery in the depropanizer?
3. What is the RVP of the C₅+ split?
4. What is the temperature difference between stages 2 and 60 of the butane splitter?

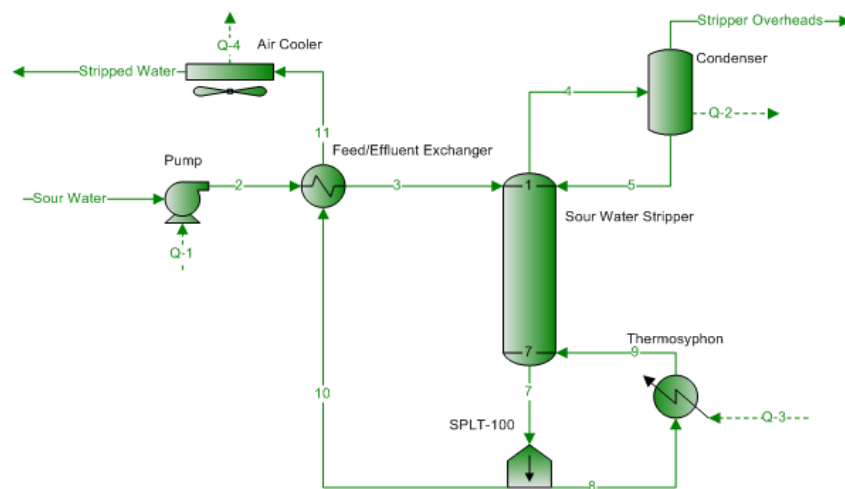
Exercise 20: SIMPLE SOUR WATER STRIPPER

This unit is used to remove several contaminants from utility water before sending it to waste water treatment. Here, we will simulate a sour water stripper with a thermosiphon and an overhead condenser.

PROCESS INFORMATION

- The pump outlet pressure is 40 psig [3.8 bar]. Use 65% pump efficiency.
- The Sour Water enters the Stripper at 190°F [90°C].
- The stripper has 7 ideal stages, and should be an Equilibrium column type.
- The top stage is 8 psig [1.5 bar] and the column has a 1 psi [0.06 bar] pressure change.
- The Condenser outlet temperature is 185 °F [93°C].
- For stripping, a thermosyphon is used to heat and return 70% of the bottoms. This stream is 15% vaporized through the exchanger.
- The Air Cooler lowers the stripped water temperature to 110°F [43°C].
- The column may require a pump-around duty estimate. Give this initial guess for the thermosyphon loop as a “Pump-around Estimate” column specification.
- After setting up the simulation, insert a Property Table to display the temperatures, pressures, flow rates, and compositions of the Sour Water, Stripped Water, Stripper Overheads streams.

Inlet Conditions	
Temperature	95°F [35°C]
Pressure	0.3psig [1.05 bar]
Flow Rate	545 sgpm [120 m ³ /h]
Composition	Mass %
NH ₃	0.55
H ₂ S	1.2
Phenol	0.1
HCN	0.001
H ₂ O	98.149



QUESTIONS

1. Determine the pump's head and required horsepower.
2. What are the end-point approach temperature, and Effective UA in the Feed Effluent Exchanger?
3. What is the Stripper bottoms temperature?
4. What is the recovery of H₂S and NH₃ in the Stripper Overhead?
5. What is the pH of the Sour Water from Tankage and the cooled Stripped Sour Water?
6. The pressure drop in the Stripper Overheads is to be maintained below 5 psi/100 ft of horizontal length. Using schedule 40 pipe, what diameter is required? What is the actual ΔP and velocity?
7. What is the critical point for the Stripper Overhead?